



American International University-Bangladesh (AIUB)

Department of Computer Science

Faculty of Science & Technology (FST)

TRADITIONAL MODEL-DRIVEN BASED BANKING SOFTWARE SYSTEM

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Table of Contents

1	PROJECT PROPOSAL	2
1.1	Background to the Problem	2
1.2	Selection of Process Model	2
2	Timeline Chart / Gantt Chart	4
3	SOFTWARE REQUIREMENTS SPECIFICATION (SRS)	5
3.1	Scopes and Features	5
3.2	User Story Table	7
3.3	Requirements Traceability Matrix	11
3.3.1	Functional Requirements	11
3.3.2	Non-Functional Requirements	15
4	SOFTWARE DESIGN	16
4.1	System Design	16
4.2	UI / Wireframe Design using Figma	18
5	GIT WORKFLOW	19
6	CONCLUSION	41

1. PROJECT PROPOSAL

1.1 Background to the Problem

Traditional banking systems are often heavily reliant on manual processes, legacy software, and decentralized data management. Bank administrators, managers, and tellers frequently face challenges in streamlining daily financial operations due to a lack of interconnected platforms. This disconnection often results in delayed transaction processing, increased risk of human error during cash handling, and prolonged wait times for customers requesting services like loans or card issuances. Furthermore, managing staff profiles and monitoring system security without an automated audit trail leaves traditional systems vulnerable to security breaches and compliance issues.

From the customer's perspective, limited access to instant financial services creates significant dissatisfaction. Customers must often visit branches physically to update basic contact information, apply for loans, or request account statements. The gap between the growing demand for instant, unified digital banking and the limitations of outdated systems requires an urgent architectural overhaul.

The proposed **Traditional Model-Driven based Banking Software System** is designed to directly address these inefficiencies. By establishing a robust, role-based architecture, the system provides an integrated platform where Admins can oversee security and staff, Managers can seamlessly process loan and card requests, and Tellers can execute physical transactions safely. Most importantly, it empowers Customers with a unified digital dashboard to manage their accounts, transfer funds securely, and track service requests in real-time. This comprehensive approach ensures a proactive, secure, and highly efficient banking environment for all stakeholders.

1.2 Selection of Process Model

For the development of this Banking Software System, the **V-Model (Verification and Validation Model)** has been selected as the primary Software Development Life Cycle (SDLC) process model.

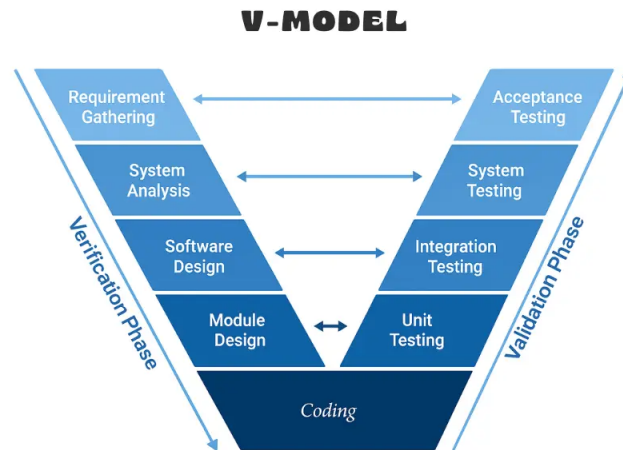


Figure 1: The V-Model Process Architecture

Why did we choose this model?

Financial and banking software systems require the highest degree of security, strict regulatory compliance, and zero tolerance for failure. The V-Model was specifically chosen because it enforces a highly disciplined approach where testing (Validation) is planned in parallel with a corresponding phase of development (Verification). 1. **Rigorous Testing:** Every stage of requirement analysis and system design has a matching testing phase (e.g., Requirements match Acceptance Testing, System Design matches System Testing), ensuring that no financial vulnerability is overlooked. 2. **Clear Traceability:** Our extensive use of Requirements Traceability Matrices (RTM) perfectly aligns with the V-Model's structure, allowing us to map every line of code back to a specific business rule. 3. **Early Defect Detection:** By preparing test plans before coding begins, architectural flaws are identified early, preventing costly and dangerous data-handling errors in the final banking application.

2. Timeline Chart / Gantt Chart

Tasks/activity	Wk 1	Wk 2	Wk 3	Wk 4	Wk 5	Wk 6	Wk 7	Wk 8	Wk 9	Wk 10	Wk 11	Wk 12
Project planning												
Requirements gathering												
SRS preparation												
USE case diagram design												
UI/wireframe design(Figma)												
Git documentation												
Trello Documentation												
Final report Writing												
Final review and submission												

Table 1: Project Timeline and Gantt Chart

3. SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

3.1 Scopes and Features

Scope 1: Profile and Account Management

- a. Bank Administrators can create and manage profiles for bank staff (Managers and Tellers).
- b. All users can log into the system using a secure, valid username and password.
- c. Customers can register for an online banking profile using their physical account number and OTP verification.
- d. After successful login, all users are redirected to their role-specific dashboard.
- e. Customers can view their account details, personal information, and current balance.
- f. Customers can update their contact information (phone number, email) with valid inputs.
- g. All users can securely change their account passwords and recover them via verification.

Scope 2: Transaction Management

- a. Tellers can process physical cash deposits and withdrawals for the customers.
- b. Customers can initiate online fund transfers to other accounts within the same bank.
- c. Customers can transfer funds to external bank accounts using valid routing information.
- d. The system automatically updates the respective account balances immediately after a transaction.
- e. Customers and Tellers can view the real-time status of any transaction (Successful/Pending/Failed).
- f. The system automatically generates a digital receipt for every successful transaction.

Scope 3: Loan and Service Management

- a. Customers can apply for different types of bank loans (e.g., Personal, Home, Education).
- b. Branch Managers can review loan applications and change the status to Approved or Rejected.
- c. Customers can track the current status of their active loan applications from their dashboard.
- d. Customers can submit requests for new debit or credit cards.

- e. Branch Managers can issue, activate, or block customer debit/credit cards based on requests.
- f. The system calculates and updates monthly loan interest automatically for active loans.

Scope 4: Reporting and Audit Management

- a. Customers can generate and download their monthly account statements in document format.
- b. Tellers can view their daily cash-in and cash-out summaries at the end of their shift.
- c. Branch Managers can view and generate branch-wise daily and monthly transaction reports.
- d. Bank Administrators can monitor active login sessions and system access logs for security.
- e. Bank Administrators can view an automated audit trail of all critical financial activities.
- f. Branch Managers can view the total number of active customers and accounts in their branch.

3.2 User Story Table

As A/An	I want to	So that	Acceptance Criteria
Bank Administrator	Create and manage profiles for bank staff (Managers and Tellers).	Authorized staff can securely access the system and perform their duties.	User must input required valid staff details, and the system saves it successfully with a notification.
Bank Administrator	Monitor active login sessions and system access logs.	I can ensure platform security and detect any unauthorized access.	The system displays an accurate list of all active sessions and recent access history.
Bank Administrator	View an automated audit trail of all critical financial activities.	I can investigate any financial discrepancies, errors, or fraud.	The audit table lists all critical actions with accurate timestamps and user IDs.
Branch Manager	Review loan applications and update their status (Approved/Rejected).	I can evaluate financial requests based on bank policies.	The application status successfully changes and notifies the customer.
Branch Manager	Issue, activate, or block customer debit/credit cards.	I can manage card services based on user requests and security needs.	The system reflects the card's active or blocked status instantly after the action.
Branch Manager	View and generate branch-wise daily and monthly transaction reports.	I can monitor the branch's overall financial performance.	Generated reports display accurate total transactions for the selected period.
Branch Manager	View the total number of active customers and accounts in my branch.	I can understand and track the branch's customer base.	A dashboard counter accurately displays the total number of active users and accounts.
Continued on next page			

Table 2 – continued from previous page

As A/An	I want to	So that	Acceptance Criteria
Bank Teller	Process physical cash deposits and withdrawals for the customers.	Customers can manage their physical funds at the branch.	The customer's account balance updates correctly based on the cash input/output.
Bank Teller	View my daily cash-in and cash-out summaries.	I can balance my cash drawer at the end of my shift.	The summary table shows the exact cash inflow and outflow for the specific shift.
Customer	Register for an online banking account using my physical account number.	I can access online banking features from my home.	The system validates the account number and OTP, allowing the user to safely set a password.
Customer	View my account details, personal information, and current balance.	I can stay informed about my financial status.	Dashboard displays an information list showing accurate profile details and balance.
Customer	Update my contact information (phone number, email).	The bank always has my most recent and valid details.	The system accepts valid new information and instantly updates the profile.
Customer	Initiate online fund transfers to other accounts within the same bank.	I can easily send money to other local accounts.	Funds are deducted from my account and added to the receiver's account instantly.
Customer	Transfer funds to external bank accounts.	I can send money to people using different banks.	The system processes the valid routing info and transfers the funds securely.
Customer	Apply for different types of bank loans (Personal, Home, Education).	I can get financial assistance when needed.	The system successfully accepts the loan application and required documents.
Continued on next page			

Table 2 – continued from previous page

As A/An	I want to	So that	Acceptance Criteria
Customer	Track the current status of my active loan applications.	I know if my loan is pending, approved, or rejected.	The dashboard shows the accurate real-time status of the applied loan.
Customer	Submit requests for new debit or credit cards.	I can get a card for physical and online transactions.	Card request is successfully submitted and pending branch manager's approval.
Customer	Generate and download my monthly account statements.	I can keep a secure record of my monthly financial activities.	A statement in document format is downloaded successfully containing all records.
User (All)	Log into the system using a secure, valid username and password.	I can securely access my account and protect my data.	After successful login, the data associated with the user should be accessible.
User (All)	Be redirected to my role-specific dashboard after successful login.	I can access tools and menus relevant only to my specific role.	Dashboard window specific to the user's role is displayed.
User (All)	Securely change my account password.	My account can be more secure.	User must input old password, new password twice, and receive a success notification.
User (All)	Recover my password via verification if I forget it.	I do not permanently lose access to my financial data.	An OTP/Link is sent to the user's registered email or phone for confirmation.
User (All)	Have the system automatically update account balances after a transaction.	The financial records remain perfectly accurate at all times.	Balance reflects the exact mathematically correct amount post-transaction.
Continued on next page			

Table 2 – continued from previous page

As A/An	I want to	So that	Acceptance Criteria
User (All)	View the real-time status of any transaction (Successful/Pending/Failed).	I know exactly whether my fund transfer or deposit was completed.	Transaction history displays correct and updated real-time statuses.
User (All)	Receive an automatically generated digital receipt for successful transactions.	I have solid proof of the transaction for future reference.	The system generates a downloadable or viewable digital receipt upon transaction success.

3.3 Requirements Traceability Matrix

3.3.1 Functional Requirements

Bank Administrator Requirements

1.0	User home page-All features availability
1.1	Login
1.2	Dashboard
1.2.1	Side menu
1.2.1.1	Dashboard button [ref 1.0]
1.2.1.2	View profile button
1.2.1.3	Edit profile information button
1.2.1.4	Change password button
1.2.2	Staff Management button
1.2.2.1	Create new Branch Manager profile form
1.2.2.2	Create new Bank Teller profile form
1.2.2.3	Submit profile creation
1.2.3	Security & Audit button
1.2.3.1	View active login sessions table
1.2.3.2	View system access logs
1.2.3.3	View automated audit trail table
1.2.4	Logout button
1.2.5	Header with User Role

Branch Manager Requirements

2.0	User home page-All features availability
2.1	Login [ref 1.1]
2.2	Dashboard
2.2.1	Side menu
2.2.1.1	Dashboard button [ref 2.0]
2.2.1.2	View profile button [ref 1.2.1.2]
2.2.1.3	Edit profile information button [ref 1.2.1.3]
2.2.1.4	Change password button [ref 1.2.1.4]
2.2.2	Loan Management button
2.2.2.1	View pending customer loan applications table
2.2.2.2	Review and click "Approve" or "Reject" status button
2.2.3	Card Service Management button
2.2.3.1	View pending debit/credit card requests
2.2.3.2	Issue, activate, or block customer card button
2.2.4	Reports & Analytics button
2.2.4.1	Generate branch-wise daily transaction report
2.2.4.2	Generate branch-wise monthly transaction report
2.2.4.3	View total active customers and accounts counter
2.2.5	Logout button [ref 1.2.4]

Bank Teller Requirements

3.0	User home page-All features availability
3.1	Login [ref 1.1]
3.2	Dashboard
3.2.1	Side menu
3.2.1.1	Dashboard button [ref 3.0]
3.2.1.2	View profile button [ref 1.2.1.2]
3.2.1.3	Edit profile information button [ref 1.2.1.3]
3.2.1.4	Change password button [ref 1.2.1.4]
3.2.2	Physical Transaction button
3.2.2.1	Search customer account by ID/Account Number
3.2.2.2	Input cash deposit amount and submit
3.2.2.3	Input cash withdrawal amount and submit
3.2.2.4	Generate digital receipt button
3.2.3	Transaction Status & Shift Summary button
3.2.3.1	View real-time status of recent transactions
3.2.3.2	View daily cash-in and cash-out summary table
3.2.4	Logout button [ref 1.2.4]

Customer Requirements

4.0	User home page-All features availability
4.1	Registration / Sign Up
4.1.1	Input Account Number and Email/Phone
4.1.2	Verify Identity via OTP

4.1.3	Set Online Password and Submit
4.2	Login [ref 1.1]
4.3	Dashboard
4.3.1	Side menu
4.3.1.1	Dashboard button (Displays Account No & Balance)
4.3.1.2	View profile button
4.3.1.3	Edit contact information (Phone/Email) button
4.3.1.4	Change password button
4.3.1.5	Recover password via verification link/OTP
4.3.2	Fund Transfer button
4.3.2.1	Initiate internal transfer (Same Bank) form
4.3.2.2	Initiate external transfer (Other Bank) form
4.3.2.3	Submit transfer request
4.3.3	Services & Loan button
4.3.3.1	Apply for Personal/Home/Education loan form
4.3.3.2	Track active loan application status
4.3.3.3	Request new debit/credit card button
4.3.4	Statements & History button
4.3.4.1	View real-time transaction history
4.3.4.2	Generate and download monthly account statement
4.3.5	Logout button [ref 1.2.4]

3.3.2 Non-Functional Requirements

5.0	System Quality Attributes
5.1	Security
5.1.1	The system must secure all user passwords using modern cryptographic hashing.
5.1.2	The system must implement strict Role-Based Access Control (RBAC) to ensure customers and tellers cannot access admin or manager dashboards.
5.1.3	All financial transactions and fund transfers must be heavily encrypted and compliant with standard banking security protocols.
5.2	Usability
5.2.1	The banking platform must feature a responsive UI, ensuring dashboards and transaction forms are accessible on desktop, tablet, and mobile devices.
5.2.2	The system should provide real-time visual feedback (e.g., success notifications, insufficient balance alerts) for user actions like fund transfers or profile updates.
5.3	Performance
5.3.1	The system automatically updates the respective account balances instantly (within 1-2 seconds) immediately after a successful transaction.
5.3.2	The system should be able to support multiple concurrent users (especially during peak banking hours) without significant degradation in speed.
5.4	Reliability & High Availability
5.4.1	The system must ensure 99.99% uptime, utilizing active-passive failover servers to prevent service interruption during hardware failures.
5.4.2	All financial records, audit trails, and customer data must be backed up securely in real-time to prevent data loss.
5.5	Regulatory Compliance
5.5.1	The system architecture must comply with PCI-DSS (Payment Card Industry Data Security Standard) for safely handling and storing debit/credit card information.

4. SOFTWARE DESIGN

4.1 System Design

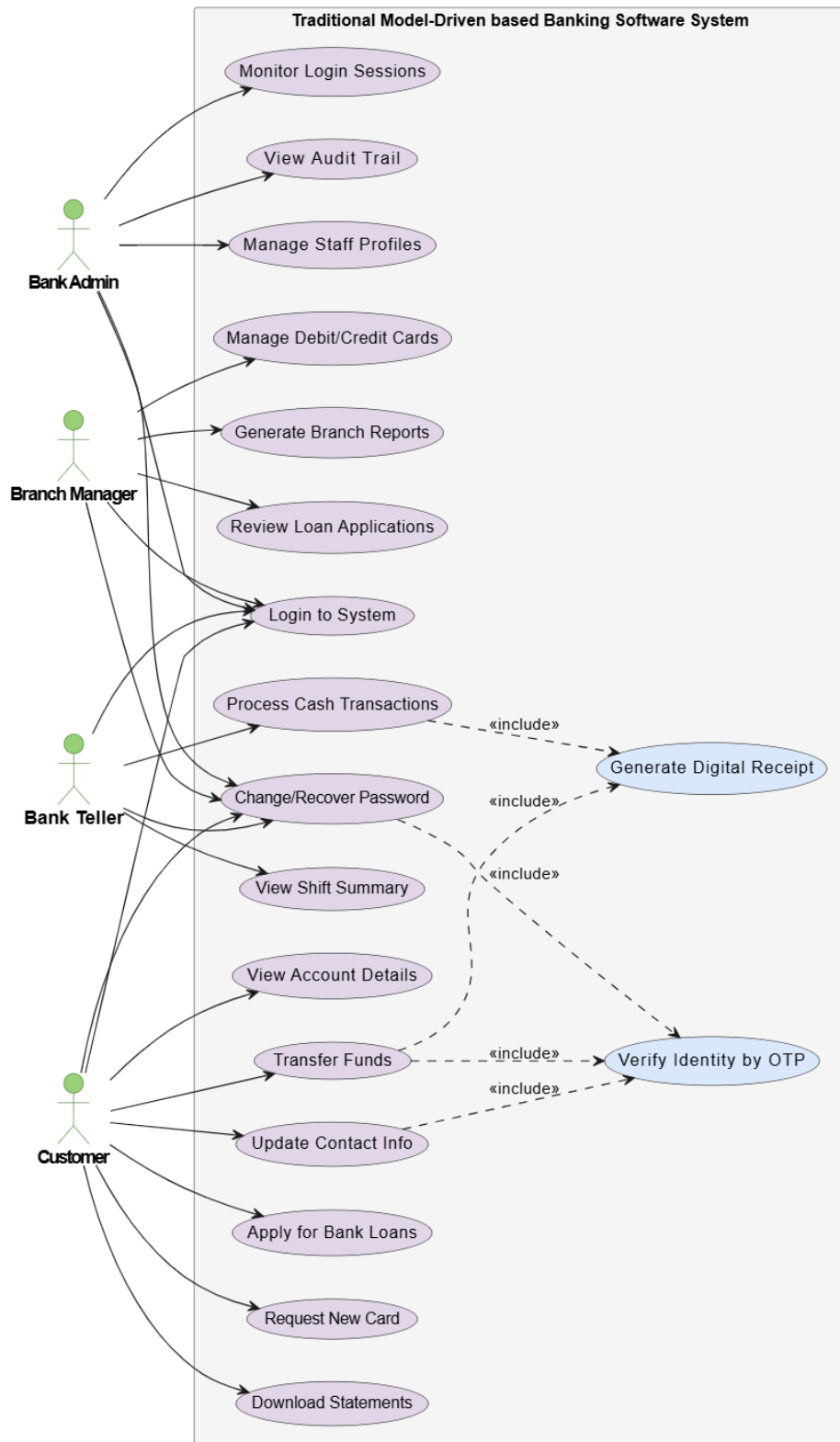


Figure 2: Use Case Diagram – Traditional Model-Driven based Banking Software System

The use case diagram represents the interaction between different types of users and the system functionalities of a banking platform developed using a structured Model-Driven approach. The system consists of four primary actors—Bank Administrator, Branch Manager, Bank Teller, and Customer—who inherit common access operations, such as secure login and OTP-based password recovery, from a general User actor. Role-specific permissions dictate that the Bank Administrator controls staff profile management, security monitoring, and automated audit trails. The Branch Manager oversees core operations by evaluating loan applications, managing card services, and generating branch-wise transaction reports. The Bank Teller handles direct physical cash deposits and withdrawals, alongside viewing daily shift summaries for balancing. Finally, the Customer is the primary end-user who can view account balances, update contact information, initiate internal and external fund transfers, apply for loans, request cards, and download monthly statements. This role-based architecture ensures efficient branch management, secure financial operations, and a seamless banking experience.

4.2 UI / Wireframe Design using Figma

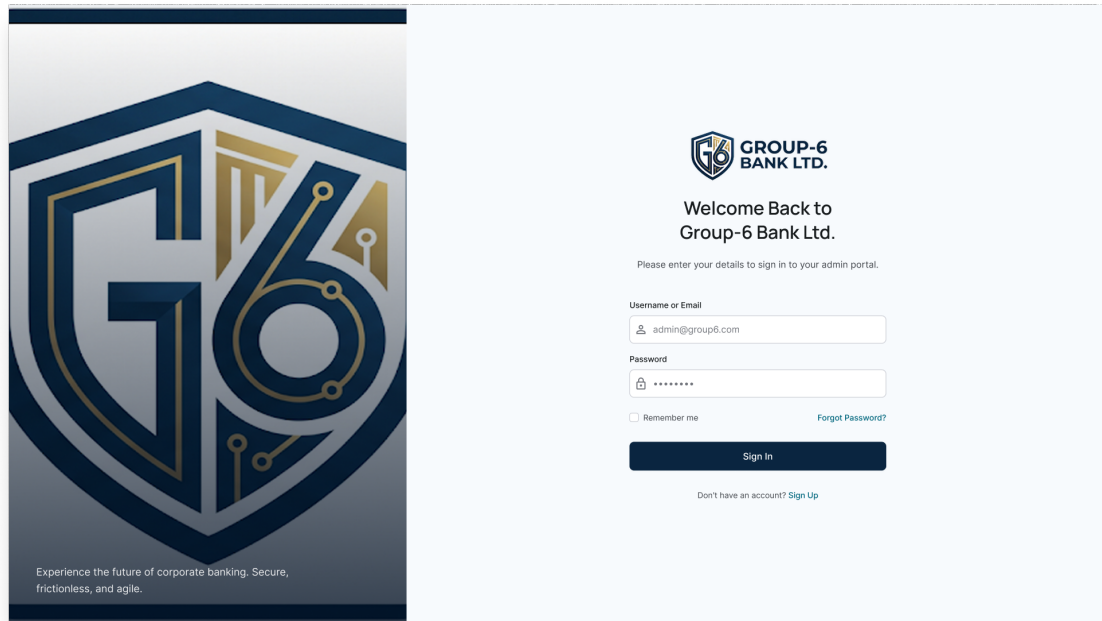


Figure 3: Figma Wireframe: Secure Login Screen

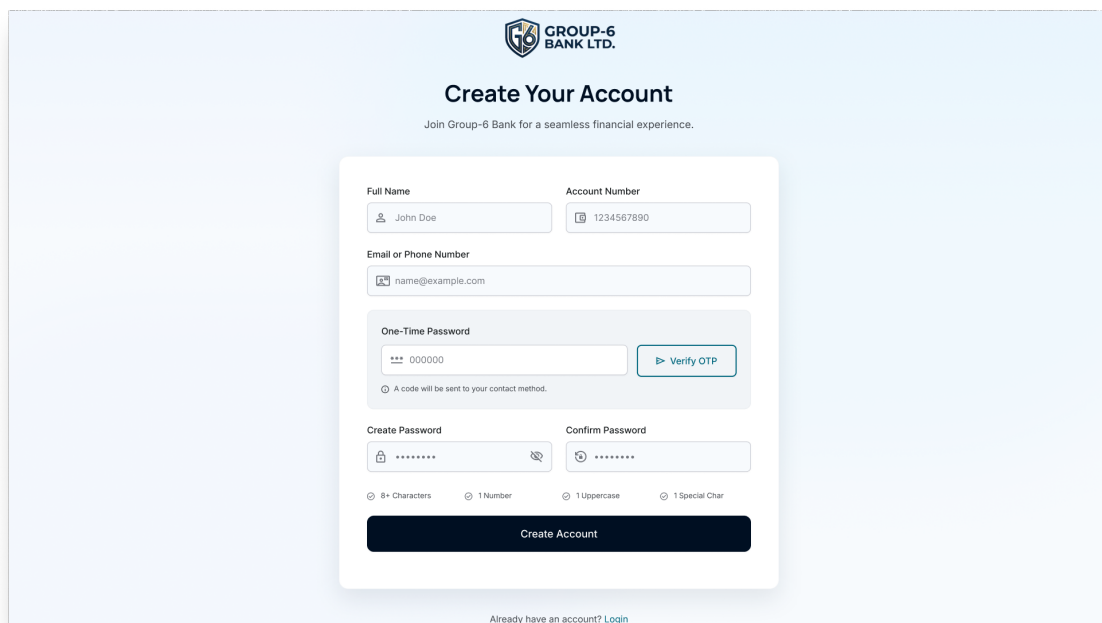


Figure 4: Figma Wireframe: Sign Up / Registration Screen

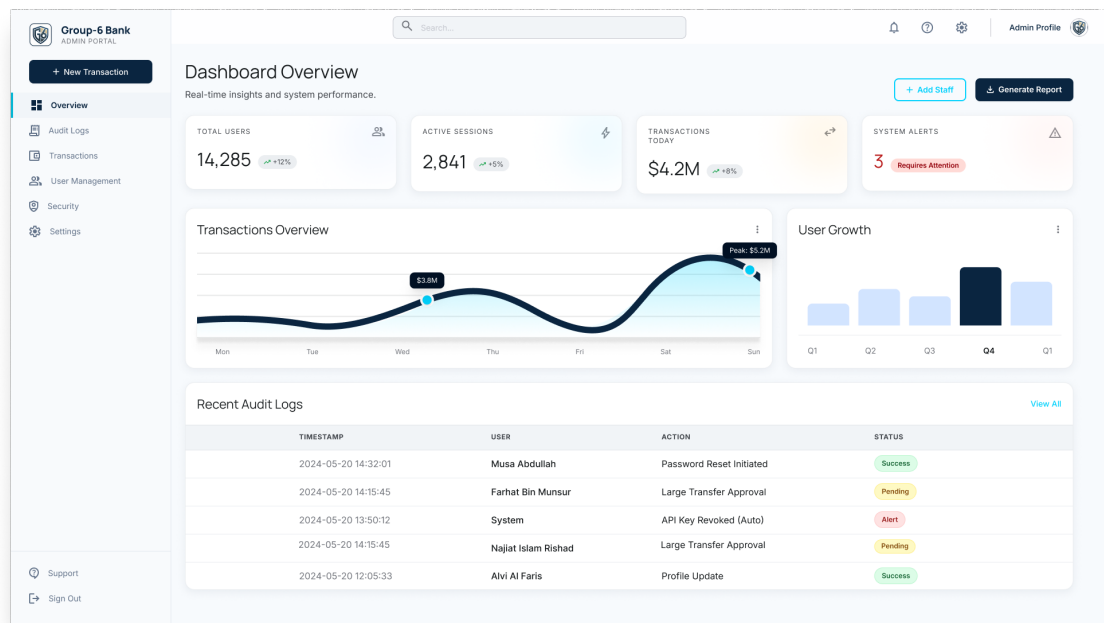


Figure 5: Figma Wireframe: Admin Dashboard Panel

5. GIT WORKFLOW

Our team established a robust version control and project management workflow utilizing Git, GitHub, and Trello. We created a central repository for the banking project on GitHub, setting the main branch as our primary integration branch. To ensure organized parallel development, each group member cloned the repository locally and created individualized feature branches for their assigned tasks based on the Trello user board.

Updates were committed with clear, descriptive messages mapping back to specific Trello cards, staged, and pushed to the remote repository. Changes were then fetched using pull commands to keep local copies updated, followed by rigorous testing and peer reviews before merging into the main branch to resolve any conflicts efficiently.

Note: The following screenshots demonstrate the detailed Git workflow, commit histories, and branch integration logs executed by different team members for their respective modules.

Admin Module Workflow

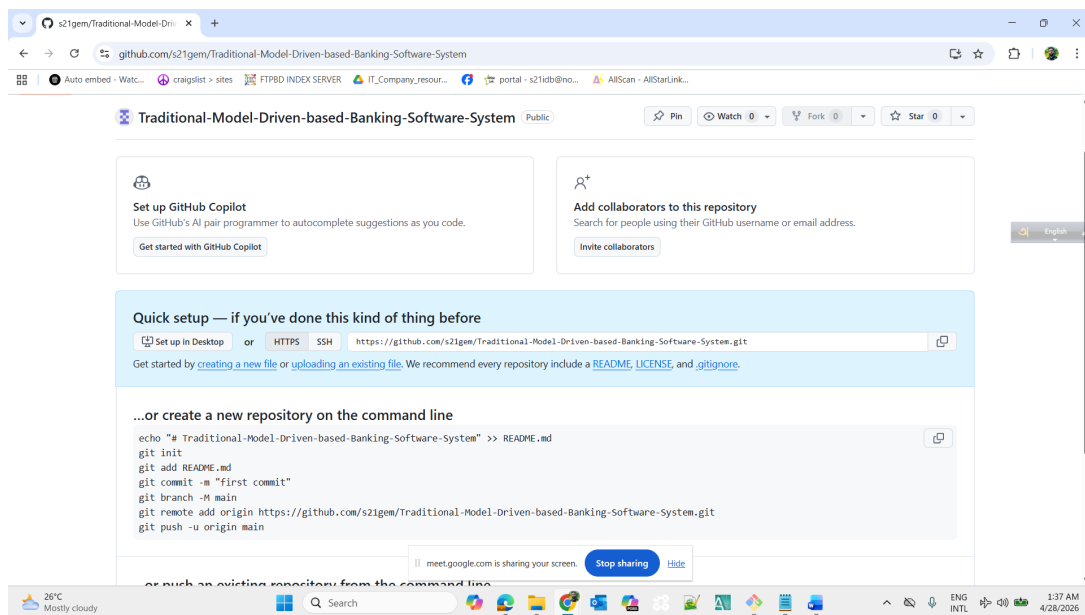


Figure 6: Admin Module Git Workflow (Part 1)

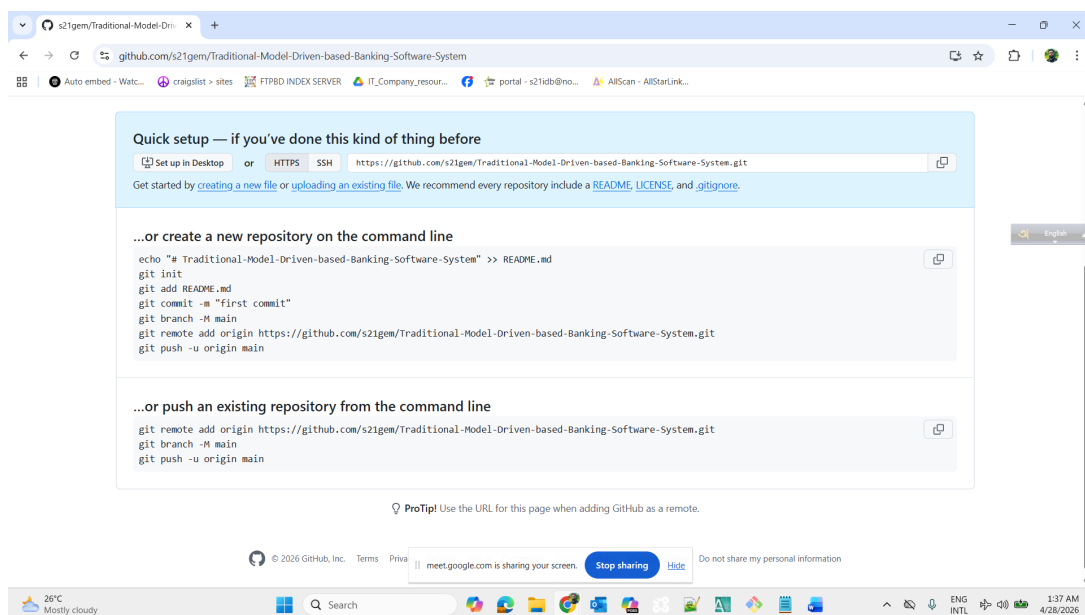
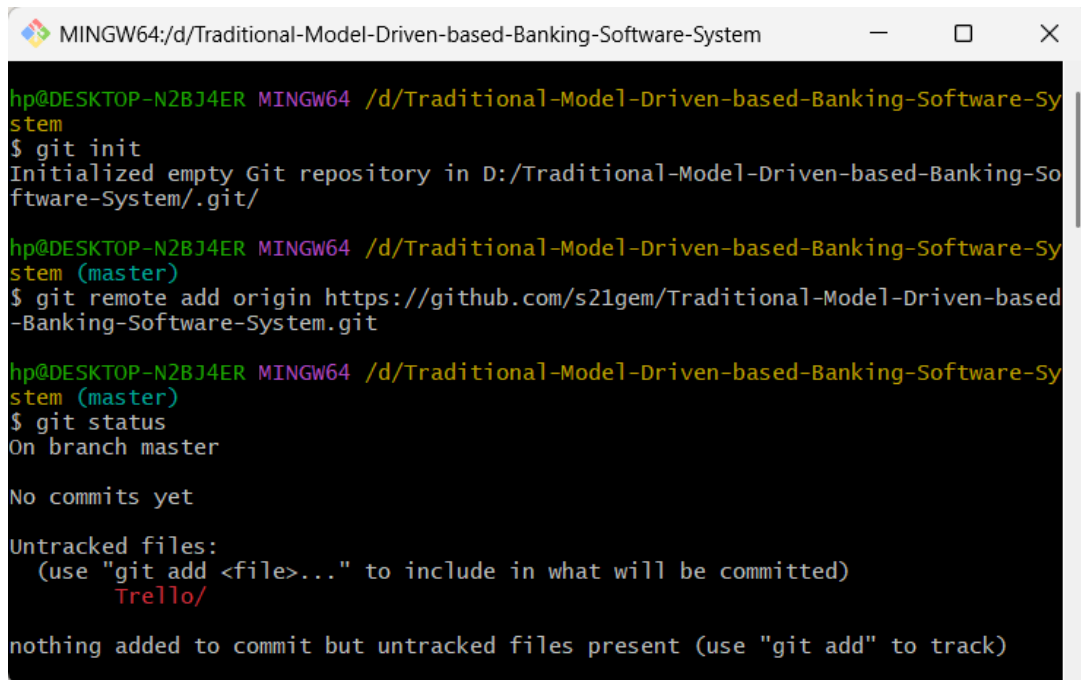


Figure 7: Admin Module Git Workflow (Part 2)



```
MINGW64:/d/Traditional-Model-Driven-based-Banking-Software-System
hp@DESKTOP-N2BJ4ER MINGW64 /d/Traditional-Model-Driven-based-Banking-Software-System
$ git init
Initialized empty Git repository in D:/Traditional-Model-Driven-based-Banking-Software-System/.git/

hp@DESKTOP-N2BJ4ER MINGW64 /d/Traditional-Model-Driven-based-Banking-Software-System (master)
$ git remote add origin https://github.com/s21gem/Traditional-Model-Driven-based-Banking-Software-System.git

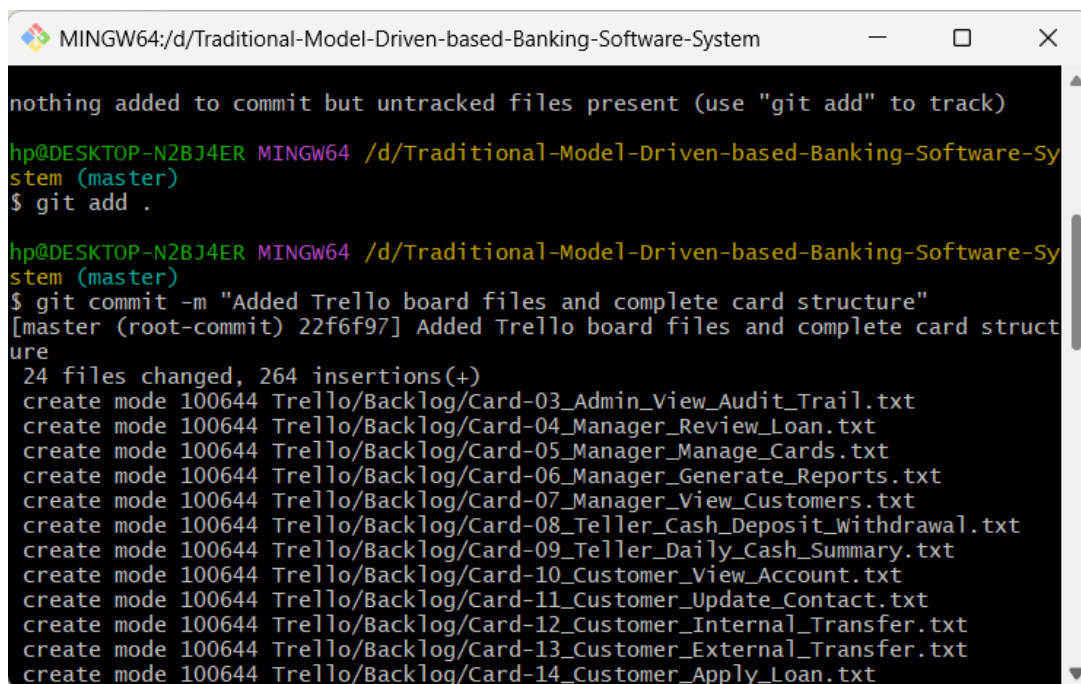
hp@DESKTOP-N2BJ4ER MINGW64 /d/Traditional-Model-Driven-based-Banking-Software-System (master)
$ git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    Trello/

nothing added to commit but untracked files present (use "git add" to track)
```

Figure 8: Admin Module Git Workflow (Part 3)



```
nothing added to commit but untracked files present (use "git add" to track)

hp@DESKTOP-N2BJ4ER MINGW64 /d/Traditional-Model-Driven-based-Banking-Software-System (master)
$ git add .

hp@DESKTOP-N2BJ4ER MINGW64 /d/Traditional-Model-Driven-based-Banking-Software-System (master)
$ git commit -m "Added Trello board files and complete card structure"
[master (root-commit) 22f6f97] Added Trello board files and complete card structure
24 files changed, 264 insertions(+)
create mode 100644 Trello/Backlog/Card-03_Admin_View_Audit_Trail.txt
create mode 100644 Trello/Backlog/Card-04_Manager_Review_Loan.txt
create mode 100644 Trello/Backlog/Card-05_Manager_Manage_Cards.txt
create mode 100644 Trello/Backlog/Card-06_Manager_Generate_Reports.txt
create mode 100644 Trello/Backlog/Card-07_Manager_View_Customers.txt
create mode 100644 Trello/Backlog/Card-08_Teller_Cash_Deposit-Withdrawal.txt
create mode 100644 Trello/Backlog/Card-09_Teller_Daily_Cash_Summary.txt
create mode 100644 Trello/Backlog/Card-10_Customer_View_Account.txt
create mode 100644 Trello/Backlog/Card-11_Customer_Update_Contact.txt
create mode 100644 Trello/Backlog/Card-12_Customer_Internal_Transfer.txt
create mode 100644 Trello/Backlog/Card-13_Customer_External_Transfer.txt
create mode 100644 Trello/Backlog/Card-14_Customer_Apply_Loan.txt
```

Figure 9: Admin Module Git Workflow (Part 4)

```
MINGW64:/d/Traditional-Model-Driven-based-Banking-Software-System
create mode 100644 Trello/Backlog/Card-13_Customer_External_Transfer.txt
create mode 100644 Trello/Backlog/Card-14_Customer_Apply_Loan.txt
create mode 100644 Trello/Backlog/Card-15_Customer_Track_Loan.txt
create mode 100644 Trello/Backlog/Card-16_Customer_Request_Card.txt
create mode 100644 Trello/Backlog/Card-17_Customer_Download_Statement.txt
create mode 100644 Trello/Backlog/Card-21_AllUsers_Recover_Password.txt
create mode 100644 Trello/Backlog/Card-22_AllUsers_Auto_Update_Balance.txt
create mode 100644 Trello/Backlog/Card-23_AllUsers_Transaction_Status.txt
create mode 100644 Trello/Backlog/Card-24_AllUsers_Digital_Receipt.txt
create mode 100644 Trello/In Progress/Card-01_Admin_Create_Staff_Profile.txt
create mode 100644 Trello/To Do/Card-02_Admin_Monitor_Login_Sessions.txt
create mode 100644 Trello/To Do/Card-18_AllUsers_Login_System.txt
create mode 100644 Trello/To Do/Card-19_AllUsers_Role_Dashboard.txt
create mode 100644 Trello/To Do/Card-20_AllUsers_Change_Password.txt

hp@DESKTOP-N2BJ4ER MINGW64 /d/Traditional-Model-Driven-based-Banking-Software-System (master)
$ git branch -M main

hp@DESKTOP-N2BJ4ER MINGW64 /d/Traditional-Model-Driven-based-Banking-Software-System (main)
$ git push -u origin main
Enumerating objects: 30, done.
Counting objects: 100% (30/30), done.
Delta compression using up to 8 threads
```

Figure 10: Admin Module Git Workflow (Part 5)

```
MINGW64:/d/Traditional-Model-Driven-based-Banking-Software-System
create mode 100644 Trello/To Do/Card-20_AllUsers_Change_Password.txt

hp@DESKTOP-N2BJ4ER MINGW64 /d/Traditional-Model-Driven-based-Banking-Software-System (master)
$ git branch -M main

hp@DESKTOP-N2BJ4ER MINGW64 /d/Traditional-Model-Driven-based-Banking-Software-System (main)
$ git push -u origin main
Enumerating objects: 30, done.
Counting objects: 100% (30/30), done.
Delta compression using up to 8 threads
Compressing objects: 100% (29/29), done.
Writing objects: 100% (30/30), 7.10 KiB | 1.77 MiB/s, done.
Total 30 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/s21gem/Traditional-Model-Driven-based-Banking-Software-System.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.

hp@DESKTOP-N2BJ4ER MINGW64 /d/Traditional-Model-Driven-based-Banking-Software-System (main)
$ git checkout -b feature-admin-module
Switched to a new branch 'feature-admin-module'
```

Figure 11: Admin Module Git Workflow (Part 6)

```
MINGW64:/d/Traditional-Model-Driven-based-Banking-Software-System
$ git checkout -b feature-admin-module
Switched to a new branch 'feature-admin-module'

hp@DESKTOP-N2BJ4ER MINGW64 /d/Traditional-Model-Driven-based-Banking-Software-System (feature-admin-module)
$ git push -u origin feature-admin-module
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'feature-admin-module' on GitHub by visiting:
remote: https://github.com/s21gem/Traditional-Model-Driven-based-Banking-Software-System/pull/new/feature-admin-module
remote:
To https://github.com/s21gem/Traditional-Model-Driven-based-Banking-Software-System.git
 * [new branch]      feature-admin-module -> feature-admin-module
branch 'feature-admin-module' set up to track 'origin/feature-admin-module'.

hp@DESKTOP-N2BJ4ER MINGW64 /d/Traditional-Model-Driven-based-Banking-Software-System (feature-admin-module)
$ ^C

hp@DESKTOP-N2BJ4ER MINGW64 /d/Traditional-Model-Driven-based-Banking-Software-System (feature-admin-module)
$
```

Figure 12: Admin Module Git Workflow (Part 7)

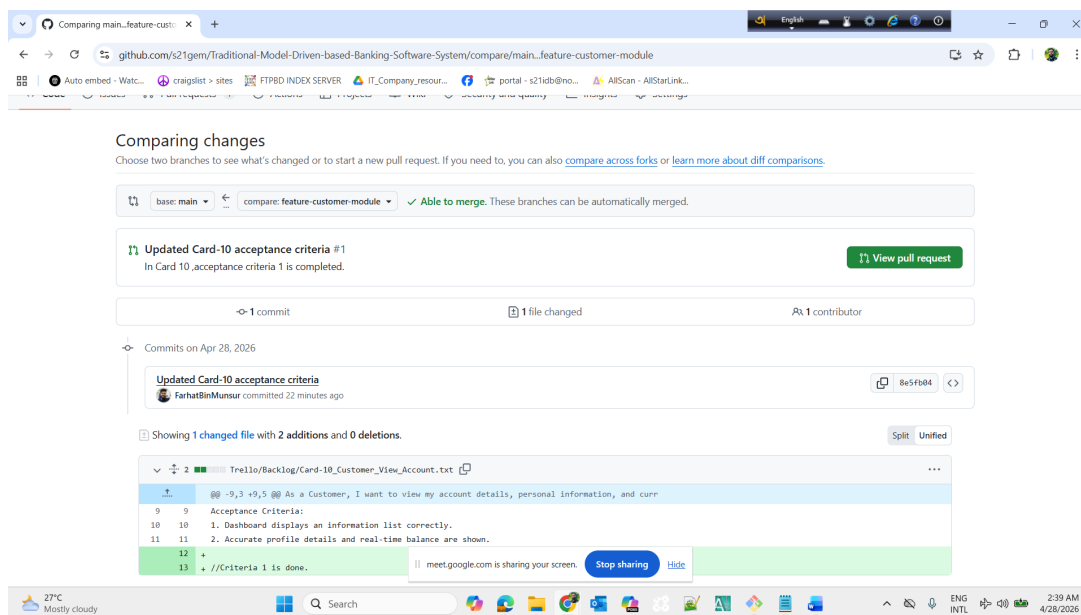


Figure 13: Admin Module Git Workflow (Part 8)

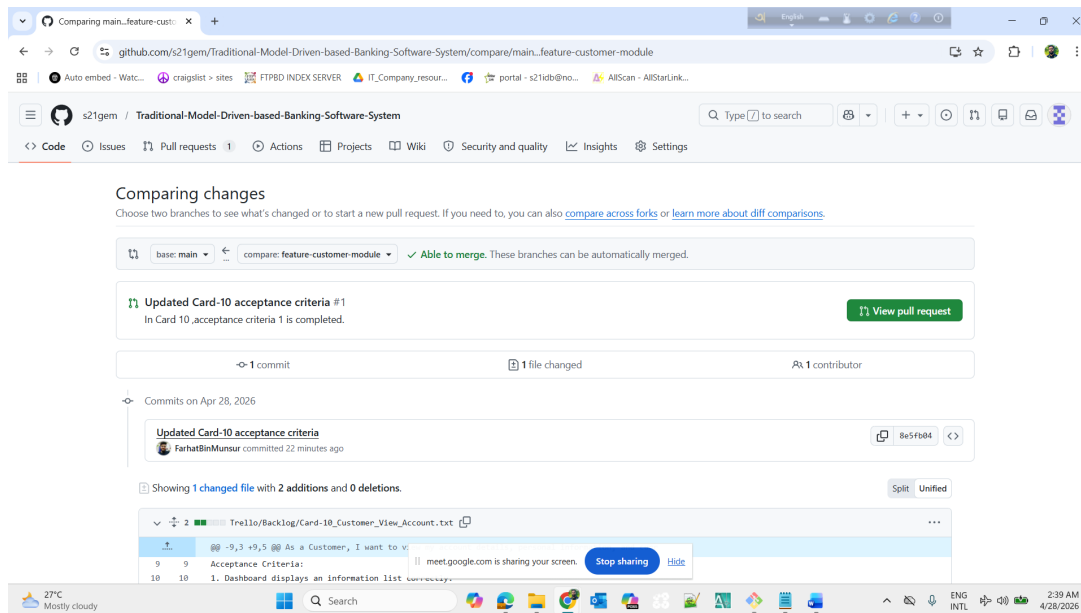


Figure 14: Admin Module Git Workflow (Part 9)

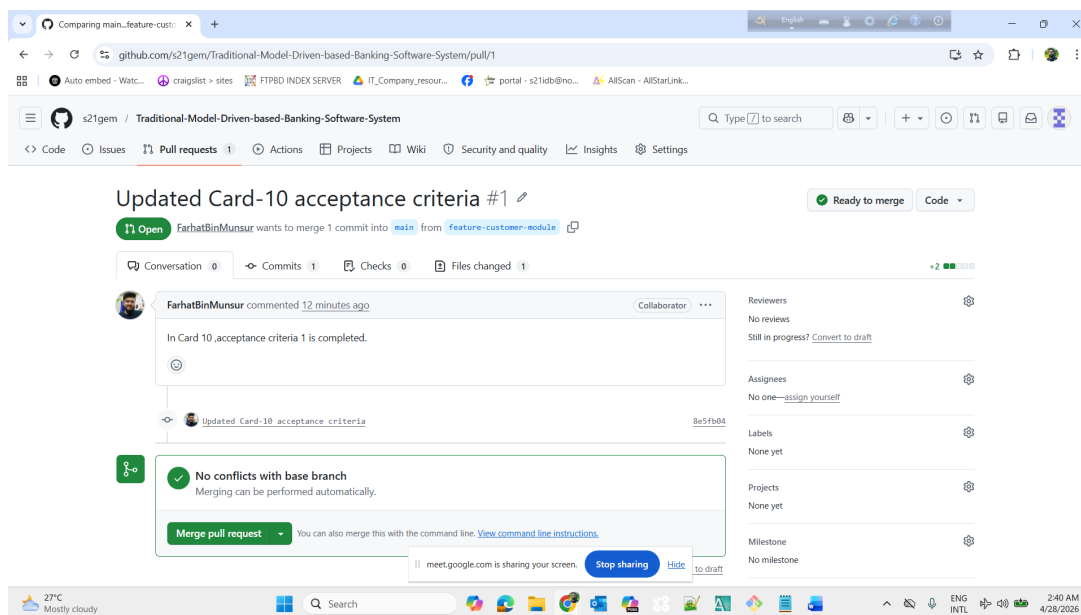


Figure 15: Admin Module Git Workflow (Part 10)

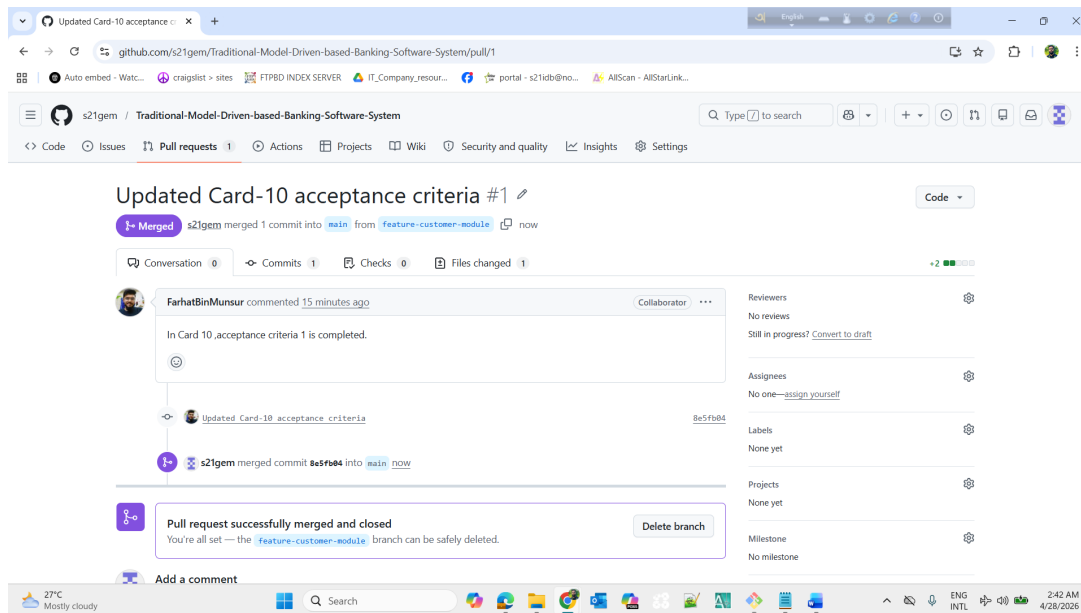


Figure 16: Admin Module Git Workflow (Part 11)

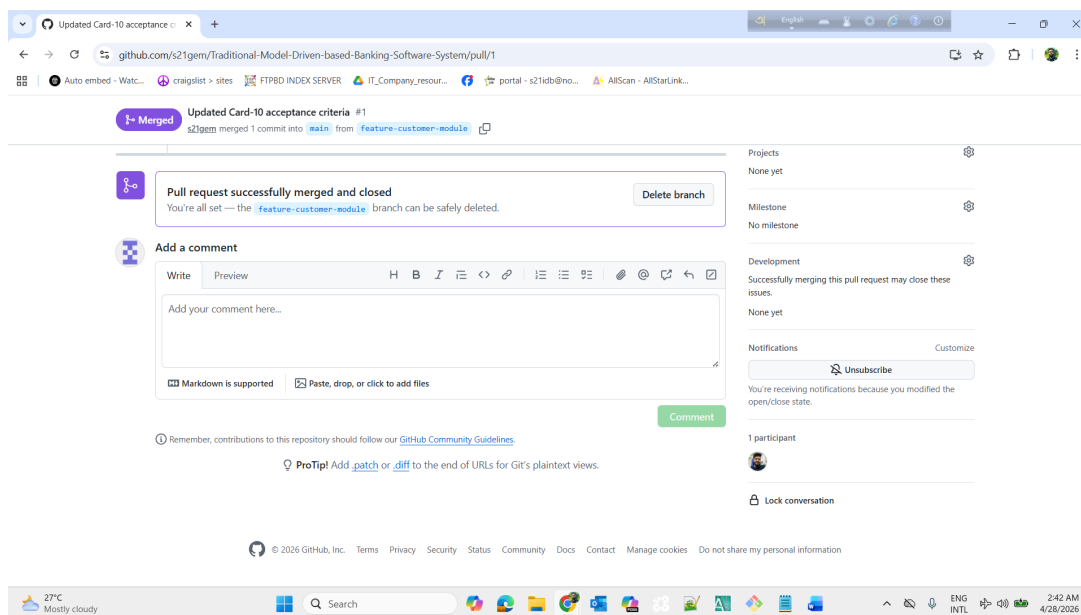


Figure 17: Admin Module Git Workflow (Part 12)

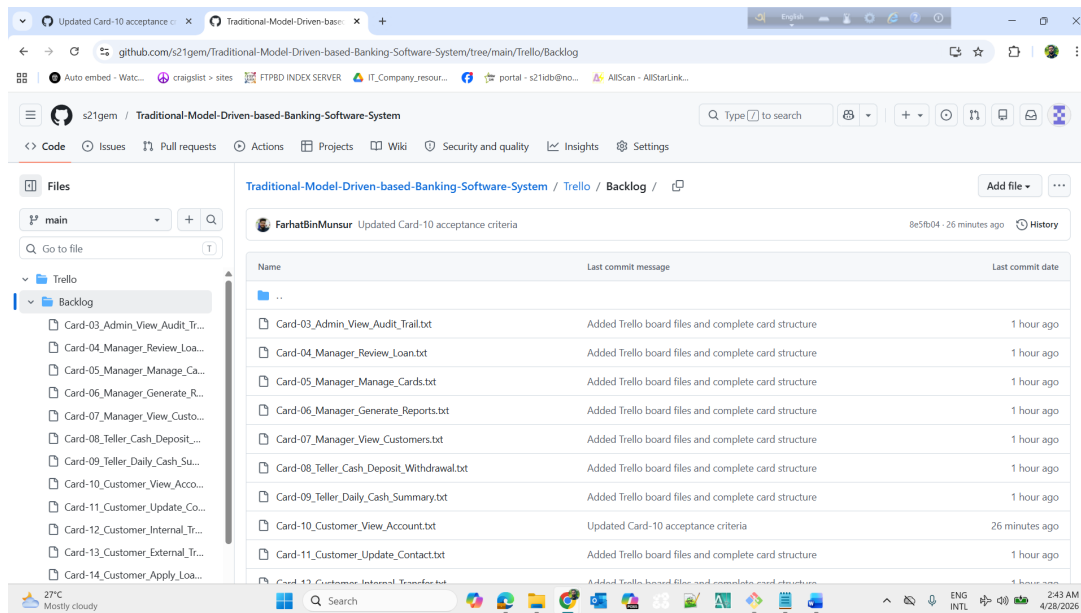


Figure 18: Admin Module Git Workflow (Part 13)

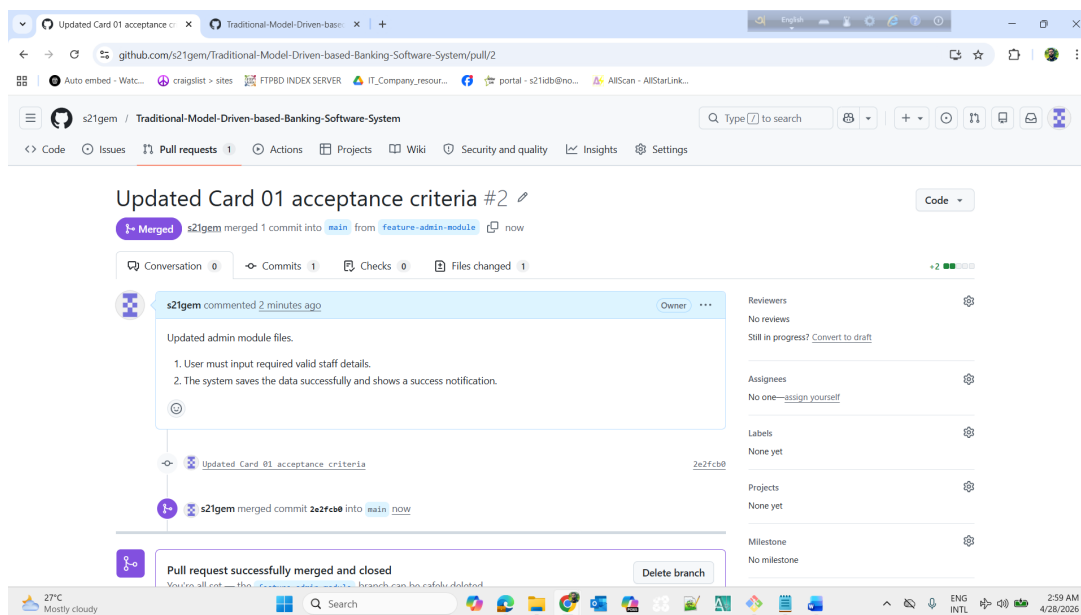


Figure 19: Admin Module Git Workflow (Part 14)

```
MINGW64:/d/Traditional-Model-Driven-based-Banking-Software-System
stem (main)
$ AC

hp@DESKTOP-N2BJ4ER MINGW64 /d/Traditional-Model-Driven-based-Banking-Software-Sy
stem (main)
$ git merge origin/feature-admin-module
Updating 22f6f97..8e5fb04
Fast-forward
 Trello/Backlog/Card-10_Customer_View_Account.txt | 2 ++
 1 file changed, 2 insertions(+)

hp@DESKTOP-N2BJ4ER MINGW64 /d/Traditional-Model-Driven-based-Banking-Software-Sy
stem (main)
$ git status
On branch main
Your branch is ahead of 'origin/main' by 1 commit.
  (use "git push" to publish your local commits)

nothing to commit, working tree clean

hp@DESKTOP-N2BJ4ER MINGW64 /d/Traditional-Model-Driven-based-Banking-Software-Sy
stem (main)
$ git push origin main
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/s21gem/Traditional-Model-Driven-based-Banking-Software-Sys
tem.git
 22f6f97..8e5fb04  main -> main

hp@DESKTOP-N2BJ4ER MINGW64 /d/Traditional-Model-Driven-based-Banking-Software-Sy
stem (main)
$ git log --oneline --graph --all
* 8e5fb04 (HEAD -> main, origin/main, origin/feature-admin-module, origin/HEA
D) Updated Card-10 acceptance criteria
* 22f6f97 (origin/feature-admin-module, feature-admin-module) Added Trello board
files and complete card structure

hp@DESKTOP-N2BJ4ER MINGW64 /d/Traditional-Model-Driven-based-Banking-Software-Sy
stem (main)
$ git log --oneline --graph --all
```

Figure 20: Admin Module Git Workflow (Part 15)

```
MINGW64:/d/Traditional-Model-Driven-based-Banking-Software-System
22f6f97..8e5fb04  main -> main

hp@DESKTOP-N2BJ4ER MINGW64 /d/Traditional-Model-Driven-based-Banking-Software-Sy
stem (main)
$ git log --oneline --graph --all
* 8e5fb04 (HEAD -> main, origin/main, origin/feature-admin-module, origin/HEA
D) Updated Card-10 acceptance criteria
* 22f6f97 (origin/feature-admin-module, feature-admin-module) Added Trello board
files and complete card structure

hp@DESKTOP-N2BJ4ER MINGW64 /d/Traditional-Model-Driven-based-Banking-Software-Sy
stem (main)
$ git log --oneline --graph --all
* 8e5fb04 (HEAD -> main, origin/main, origin/feature-admin-module, origin/HEA
D) Updated Card-10 acceptance criteria
* 22f6f97 (origin/feature-admin-module, feature-admin-module) Added Trello board
files and complete card structure

hp@DESKTOP-N2BJ4ER MINGW64 /d/Traditional-Model-Driven-based-Banking-Software-System (main)
$ AC

hp@DESKTOP-N2BJ4ER MINGW64 /d/Traditional-Model-Driven-based-Banking-Software-System (main)
$ git checkout feature-admin-module
Switched to branch 'feature-admin-module'
Your branch is up to date with 'origin/feature-admin-module'.

hp@DESKTOP-N2BJ4ER MINGW64 /d/Traditional-Model-Driven-based-Banking-Software-System (feature-admin-module)
$ git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.

hp@DESKTOP-N2BJ4ER MINGW64 /d/Traditional-Model-Driven-based-Banking-Software-System (main)
$ git pull origin main
From https://github.com/s21gem/Traditional-Model-Driven-based-Banking-Software-System
* branch      main      -> FETCH_HEAD
Already up to date.

hp@DESKTOP-N2BJ4ER MINGW64 /d/Traditional-Model-Driven-based-Banking-Software-System (main)
$ git checkout feature-admin-module
Switched to branch 'feature-admin-module'
Your branch is up to date with 'origin/feature-admin-module'.
```

Figure 21: Admin Module Git Workflow (Part 16)


```
MINGW64:/d/Traditional-Model-Driven-based-Banking-Software-System
Already up to date.

hp@DESKTOP-N2BJ4ER MINGW64 /d/Traditional-Model-Driven-based-Banking-Software-System (main)
$ git checkout feature-admin-module
Switched to branch 'feature-admin-module'
Your branch is up to date with 'origin/feature-admin-module'.

hp@DESKTOP-N2BJ4ER MINGW64 /d/Traditional-Model-Driven-based-Banking-Software-System (feature-admin-module)
$ git merge main
Updating 22f6f97..8e5fb04
Fast-forward
 Trello/Backlog/Card-10_Customer_View_Account.txt | 2 ++
 1 file changed, 2 insertions(+)

hp@DESKTOP-N2BJ4ER MINGW64 /d/Traditional-Model-Driven-based-Banking-Software-System (feature-admin-module)
$ git status
On branch feature-admin-module
Your branch is ahead of 'origin/feature-admin-module' by 1 commit.
(use "git push" to publish your local commits)

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   Trello/In Progress/Card-01_Admin_Create_Staff_Profile.txt

no changes added to commit (use "git add" and/or "git commit -a")

hp@DESKTOP-N2BJ4ER MINGW64 /d/Traditional-Model-Driven-based-Banking-Software-System (feature-admin-module)
$ git add .

hp@DESKTOP-N2BJ4ER MINGW64 /d/Traditional-Model-Driven-based-Banking-Software-System (feature-admin-module)
$ git commit -m "Updated Card 01 acceptance criteria"
[feature-admin-module 2e2fcb0] Updated Card 01 acceptance criteria
1 file changed, 2 insertions(+)

hp@DESKTOP-N2BJ4ER MINGW64 /d/Traditional-Model-Driven-based-Banking-Software-System (feature-admin-module)
$ git push origin feature-admin-module
Enumerating objects: 9, done.
Counting objects: 100% (9/9), done.
```

Figure 22: Admin Module Git Workflow (Part 17)

```
MINGW64:/d/Traditional-Model-Driven-based-Banking-Software-System
$ git add .

hp@DESKTOP-N2BJ4ER MINGW64 /d/Traditional-Model-Driven-based-Banking-Software-System (feature-admin-module)
$ git commit -m "Updated Card 01 acceptance criteria"
[feature-admin-module 2e2fcb0] Updated Card 01 acceptance criteria
1 file changed, 2 insertions(+)

hp@DESKTOP-N2BJ4ER MINGW64 /d/Traditional-Model-Driven-based-Banking-Software-System (feature-admin-module)
$ git push origin feature-admin-module
Enumerating objects: 9, done.
Counting objects: 100% (9/9), done.
Delta compression using up to 8 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (5/5), 493 bytes | 493.00 KiB/s, done.
Total 5 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/s21gem/Traditional-Model-Driven-based-Banking-Software-System.git
 22f6f97..2e2fcb0 feature-admin-module -> feature-admin-module

hp@DESKTOP-N2BJ4ER MINGW64 /d/Traditional-Model-Driven-based-Banking-Software-System (feature-admin-module)
$ git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.

hp@DESKTOP-N2BJ4ER MINGW64 /d/Traditional-Model-Driven-based-Banking-Software-System (main)
$ git pull origin main
From https://github.com/s21gem/Traditional-Model-Driven-based-Banking-Software-System
 * branch      main      -> FETCH_HEAD
Already up to date.

hp@DESKTOP-N2BJ4ER MINGW64 /d/Traditional-Model-Driven-based-Banking-Software-System (main)
$ git fetch --all

hp@DESKTOP-N2BJ4ER MINGW64 /d/Traditional-Model-Driven-based-Banking-Software-System (main)
$ git merge origin/feature-admin-module
Updating 8e5fb04..2e2fcb0
Fast-forward
 Trello/In Progress/Card-01_Admin_Create_Staff_Profile.txt | 2 ++
 1 file changed, 2 insertions(+)


```

Figure 23: Admin Module Git Workflow (Part 18)

```

MINGW64:/d/Traditional-Model-Driven-based-Banking-Software-System
to https://github.com/s21gem/Traditional-Model-Driven-based-Banking-Software-System.git
22f6f97..2e2fcb0 feature-admin-module -> feature-admin-module

hp@DESKTOP-N2BJ4ER MINGW64 /d/Traditional-Model-Driven-based-Banking-Software-System (feature-admin-module)
$ git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.

hp@DESKTOP-N2BJ4ER MINGW64 /d/Traditional-Model-Driven-based-Banking-Software-System (main)
$ git pull origin main
From https://github.com/s21gem/Traditional-Model-Driven-based-Banking-Software-System
* branch      main      -> FETCH_HEAD
Already up to date.

hp@DESKTOP-N2BJ4ER MINGW64 /d/Traditional-Model-Driven-based-Banking-Software-System (main)
$ git fetch --all

hp@DESKTOP-N2BJ4ER MINGW64 /d/Traditional-Model-Driven-based-Banking-Software-System (main)
$ git merge origin/feature-admin-module
Updating 8e5fb04..2e2fcb0
Fast-forward
 Trello/In_Progress/Card-01_Admin_Create_Staff_Profile.txt | 2 ++
1 file changed, 2 insertions(+)

hp@DESKTOP-N2BJ4ER MINGW64 /d/Traditional-Model-Driven-based-Banking-Software-System (main)
$ git push origin main
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/s21gem/Traditional-Model-Driven-based-Banking-Software-System.git
8e5fb04..2e2fcb0 main -> main

hp@DESKTOP-N2BJ4ER MINGW64 /d/Traditional-Model-Driven-based-Banking-Software-System (main)
$ git log --oneline --graph --all
* 2e2fcb0 (HEAD -> main, origin/main, origin/feature-admin-module, origin/HEAD, feature-admin-module) Updated Card 01 acceptance criteria
* 8e5fb04 (origin/feature-customer-module) Updated Card-10 acceptance criteria
* 22f6f97 Added Trello board files and complete card structure

hp@DESKTOP-N2BJ4ER MINGW64 /d/Traditional-Model-Driven-based-Banking-Software-System (main)
$

```

Figure 24: Admin Module Git Workflow (Part 19)

```

MINGW64:/d/Traditional-Model-Driven-based-Banking-Software-System

hp@DESKTOP-N2BJ4ER MINGW64 /d/Traditional-Model-Driven-based-Banking-Software-System (main)
$ git checkout main
Already on 'main'
Your branch is up to date with 'origin/main'.

hp@DESKTOP-N2BJ4ER MINGW64 /d/Traditional-Model-Driven-based-Banking-Software-System (main)
$ git pull origin main
remote: Enumerating objects: 10, done.
remote: Counting objects: 100% (10/10), done.
remote: Compressing objects: 100% (3/3), done.
remote: Total 6 (delta 2), reused 5 (delta 2), pack-reused 0 (from 0)
Unpacking objects: 100% (6/6), 1.33 KiB | 113.00 KiB/s, done.
From https://github.com/s21gem/Traditional-Model-Driven-based-Banking-Software-System
* branch      main      -> FETCH_HEAD
* 2e2fcb0..051a200 main -> origin/main
Updating 2e2fcb0..051a200
Fast-forward
 Trello/Backlog/Card-04_Manager_Review_Loan.txt | 2 ++
1 file changed, 2 insertions(+)

hp@DESKTOP-N2BJ4ER MINGW64 /d/Traditional-Model-Driven-based-Banking-Software-System (main)
$ git fetch --all
From https://github.com/s21gem/Traditional-Model-Driven-based-Banking-Software-System
* [new branch] feature-manager-module -> origin/feature-manager-module

hp@DESKTOP-N2BJ4ER MINGW64 /d/Traditional-Model-Driven-based-Banking-Software-System (main)
$ git merge origin/feature-manager-module
Already up to date.

hp@DESKTOP-N2BJ4ER MINGW64 /d/Traditional-Model-Driven-based-Banking-Software-System (main)
$ git push origin main
Everything up-to-date

hp@DESKTOP-N2BJ4ER MINGW64 /d/Traditional-Model-Driven-based-Banking-Software-System (main)
$ git log --oneline --graph --all
* 051a200 (HEAD -> main, origin/main, origin/HEAD) Merge pull request #3 from s21gem/feature-manager-module
/*
/* c02e045 (origin/feature-manager-module) Updated Card-04 Criteria 1.
/*
* 2e2fcb0 (origin/feature-admin-module, feature-admin-module) Updated Card 01 acceptance criteria
* 8e5fb04 (origin/feature-customer-module) Updated Card-10 acceptance criteria
* 22f6f97 Added Trello board files and complete card structure

hp@DESKTOP-N2BJ4ER MINGW64 /d/Traditional-Model-Driven-based-Banking-Software-System (main)
$ ^C

hp@DESKTOP-N2BJ4ER MINGW64 /d/Traditional-Model-Driven-based-Banking-Software-System (main)
$ git pull origin main
From https://github.com/s21gem/Traditional-Model-Driven-based-Banking-Software-System
* branch      main      -> FETCH_HEAD
Already up to date.

hp@DESKTOP-N2BJ4ER MINGW64 /d/Traditional-Model-Driven-based-Banking-Software-System (main)
$ git fetch --all
remote: Enumerating objects: 9, done.

```

Figure 25: Admin Module Git Workflow (Part 20)

```
MINGW64/d/Traditional-Model-Driven-based-Banking-Software-System
Everything up-to-date

hp@DESKTOP-N2BJ4ER MINGW64 /d/Traditional-Model-Driven-based-Banking-Software-System (main)
$ git log --oneline --graph --all
* 051a200 (HEAD -> main, origin/main, origin/HEAD) Merge pull request #3 from s21gem/feature-manager-
module
/* c02e045 (origin/feature-manager-module) Updated Card-04 Criteria 1.
/* 2e2fcb0 (origin/feature-admin-module, feature-admin-module) Updated Card 01 acceptance criteria
/* 8e5fb04 (origin/feature-customer-module) Updated Card-10 acceptance criteria
/* 22f6f97 Added Trello board files and complete card structure

hp@DESKTOP-N2BJ4ER MINGW64 /d/Traditional-Model-Driven-based-Banking-Software-System (main)
$ AC

hp@DESKTOP-N2BJ4ER MINGW64 /d/Traditional-Model-Driven-based-Banking-Software-System (main)
$ git pull origin main
From https://github.com/s21gem/Traditional-Model-Driven-based-Banking-Software-System
* branch      main      -> FETCH_HEAD
Already up to date.

hp@DESKTOP-N2BJ4ER MINGW64 /d/Traditional-Model-Driven-based-Banking-Software-System (main)
$ git fetch --all
remote: Enumerating objects: 9, done.
remote: Counting objects: 100% (9/9), done.
remote: Compressing objects: 100% (1/1), done.
remote: Total 5 (delta 3), reused 5 (delta 3), pack-reused 0 (from 0)
Unpacking objects: 100% (5/5), 417 bytes | 34.00 KiB/s, done.
From https://github.com/s21gem/Traditional-Model-Driven-based-Banking-Software-System
* [new branch]      feature-general-module -> origin/feature-general-module

hp@DESKTOP-N2BJ4ER MINGW64 /d/Traditional-Model-Driven-based-Banking-Software-System (main)
$ git merge origin/feature-general-module
Updating 051a200..d4d8ec9
Fast-forward
 Trello/To Do/Card-18_AllUsers_Login_System.txt | 2 ++
 1 file changed, 2 insertions(+)

hp@DESKTOP-N2BJ4ER MINGW64 /d/Traditional-Model-Driven-based-Banking-Software-System (main)
$ git push origin main
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/s21gem/Traditional-Model-Driven-based-Banking-Software-System.git
 051a200..d4d8ec9  main -> main

hp@DESKTOP-N2BJ4ER MINGW64 /d/Traditional-Model-Driven-based-Banking-Software-System (main)
$ git log --oneline --graph --all
* d4d8ec9 (HEAD -> main, origin/main, origin/feature-general-module, origin/HEAD) Updated Criteria 1 and 2 in card 18
/* 051a200 Merge pull request #3 from s21gem/feature-manager-module
/* c02e045 (origin/feature-manager-module) Updated Card-04 Criteria 1.
/* 2e2fcb0 (origin/feature-admin-module, feature-admin-module) Updated Card 01 acceptance criteria
/* 8e5fb04 (origin/feature-customer-module) Updated Card-10 acceptance criteria
/* 22f6f97 Added Trello board files and complete card structure

hp@DESKTOP-N2BJ4ER MINGW64 /d/Traditional-Model-Driven-based-Banking-Software-System (main)
$ |
```

Figure 26: Admin Module Git Workflow (Part 21)

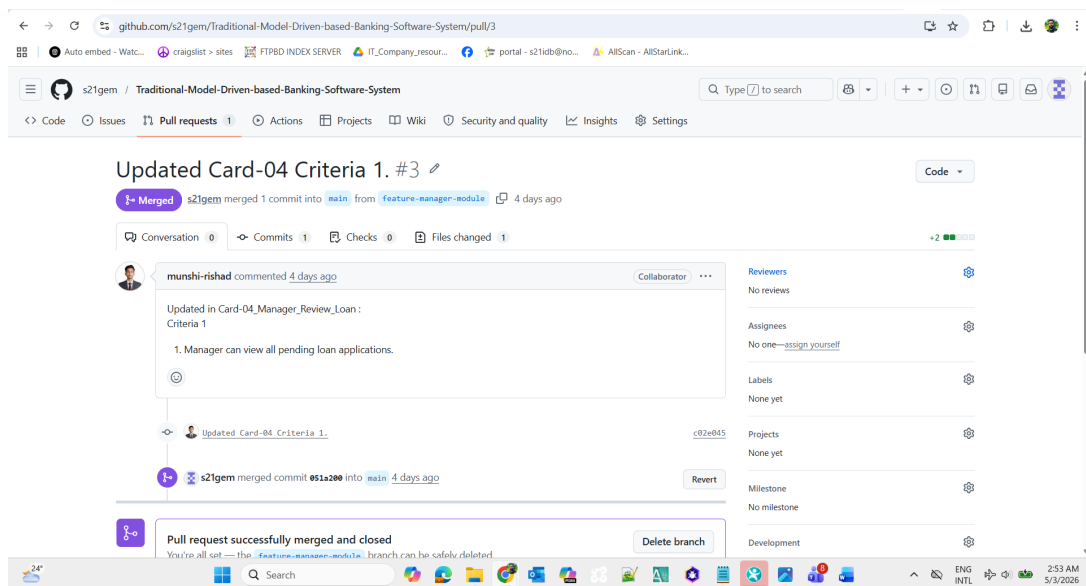


Figure 27: Admin Module Git Workflow (Part 22)

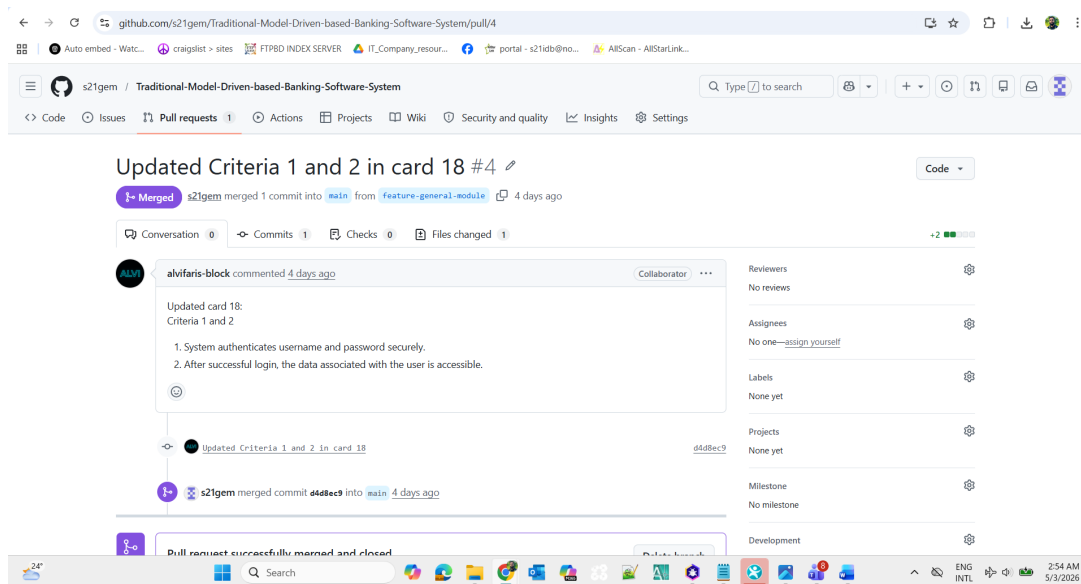


Figure 28: Admin Module Git Workflow (Part 23)

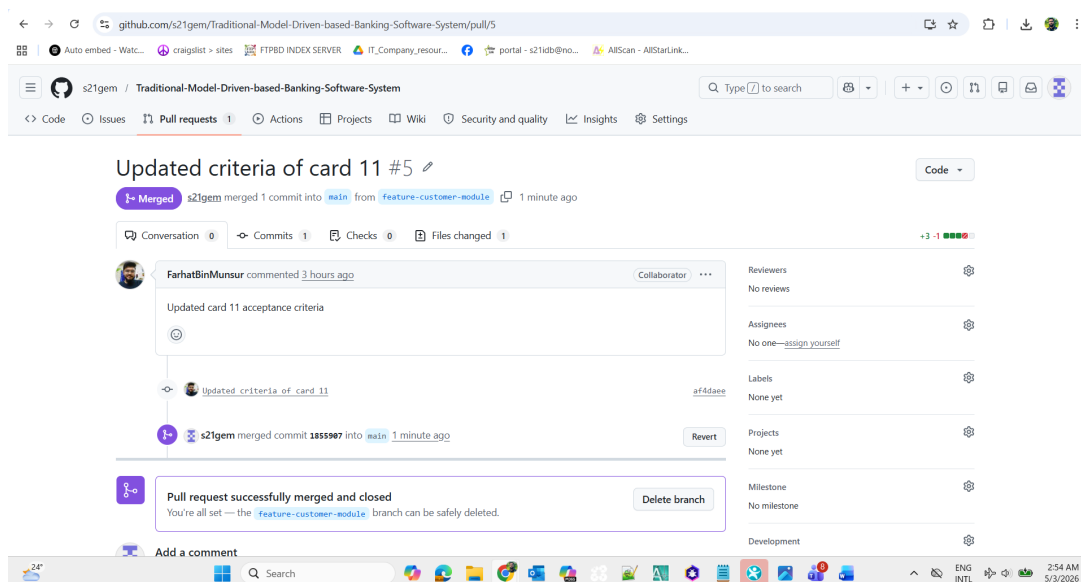


Figure 29: Admin Module Git Workflow (Part 24)

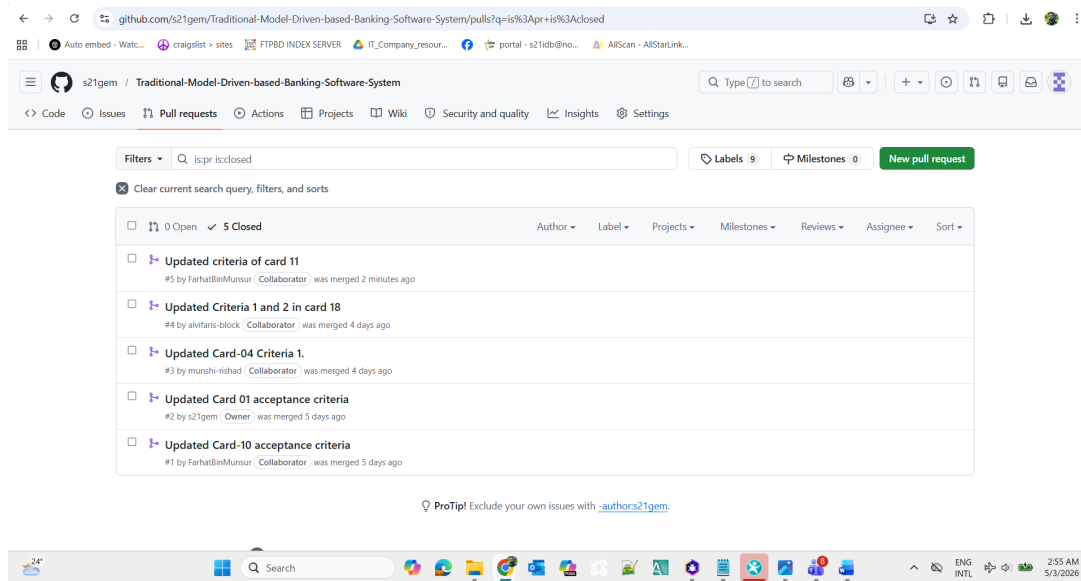


Figure 30: Admin Module Git Workflow (Part 25)

```

MINGW64:/d/Traditional-Model-Driven-based-Banking-Software-System
$ git push origin main
Everything up-to-date

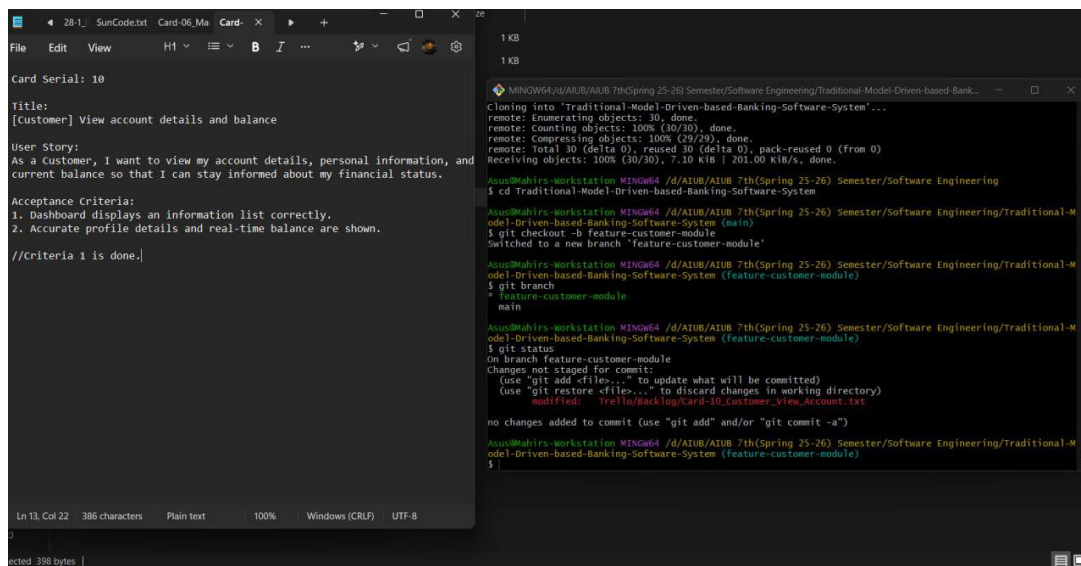
hp@DESKTOP-N2BJ4ER MINGW64 /d/Traditional-Model-Driven-based-Banking-Software-Sy
stem (main)
$ git log --oneline --graph --all
* 1855907 (HEAD -> main, origin/main, origin/HEAD) Merge pull request #5 from
s21gem/feature-customer-module
|
| * af4daee (origin/feature-customer-module) Updated criteria of card 11
|/
* d4d8ec9 (origin/feature-general-module) Updated Criteria 1 and 2 in card 18
* 051a200 Merge pull request #3 from s21gem/feature-manager-module
|
| * c02e045 (origin/feature-manager-module) Updated Card-04 Criteria 1.
|/
* 2e2fcb0 (origin/feature-admin-module, feature-admin-module) Updated Card 01 ac
ceptance criteria
* 8e5fb04 Updated Card-10 acceptance criteria
* 22f6f97 Added Trello board files and complete card structure

hp@DESKTOP-N2BJ4ER MINGW64 /d/Traditional-Model-Driven-based-Banking-Software-Sy
stem (main)
$

```

Figure 31: Admin Module Git Workflow (Part 26)

Customer Module Workflow



```
Card Serial: 10

Title:
[Customer] View account details and balance

User Story:
As a customer, I want to view my account details, personal information, and current balance so that I can stay informed about my financial status.

Acceptance Criteria:
1. Dashboard displays an information list correctly.
2. Accurate profile details and real-time balance are shown.

//Criteria 1 is done.
```

```
Cloning into 'Traditional-Model-Driven-based-Banking-Software-System'...
remote: Enumerating objects: 30, done.
remote: Counting objects: 100% (30/30), done.
remote: Compressing objects: 100% (29/29), done.
remote: Total 30 (delta 0), reused 30 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (30/30), 7.10 KiB | 201.00 KiB/s, done.

Asus@Mahirs-Workstation MINGW64 /d/AIUB/AIUB 7th(Spring 25-26) Semester/Software Engineering
$ cd Traditional-Model-Driven-based-Banking-Software-System

Asus@Mahirs-Workstation MINGW64 /d/AIUB/AIUB 7th(Spring 25-26) Semester/Software Engineering/Traditional-M
odel-Driven-based-Banking-Software-System (main)
$ git checkout -b feature-customer-module
Switched to a new branch 'feature-customer-module'

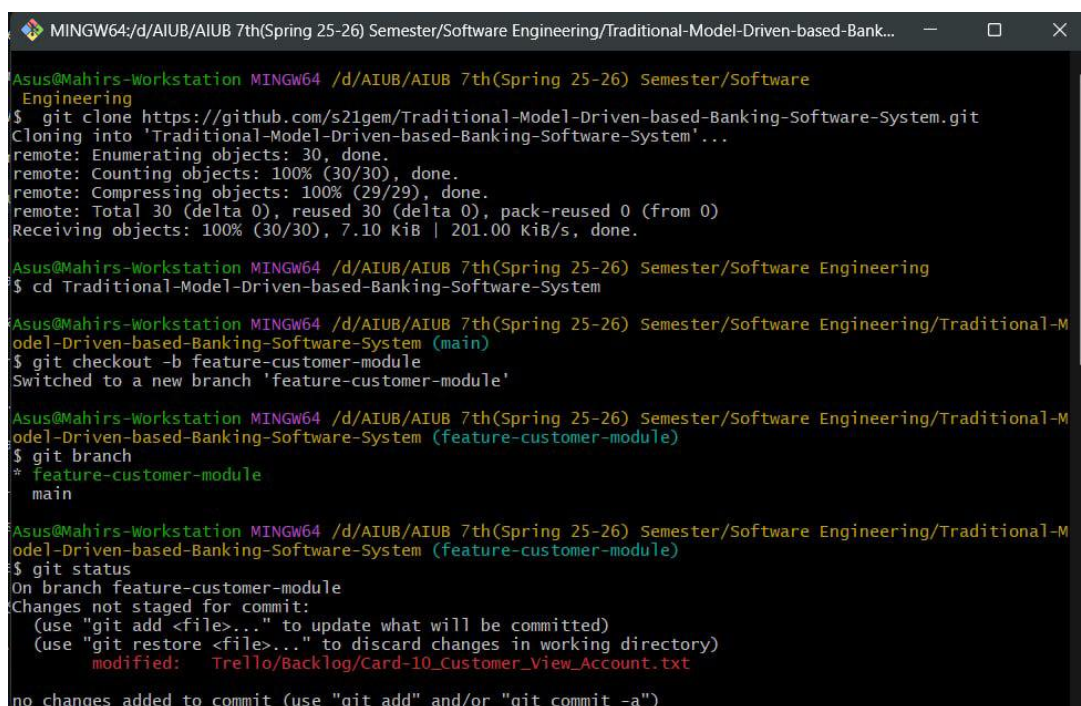
Asus@Mahirs-Workstation MINGW64 /d/AIUB/AIUB 7th(Spring 25-26) Semester/Software Engineering/Traditional-M
odel-Driven-based-Banking-Software-System (feature-customer-module)
$ git branch
* feature-customer-module
main

Asus@Mahirs-Workstation MINGW64 /d/AIUB/AIUB 7th(Spring 25-26) Semester/Software Engineering/Traditional-M
odel-Driven-based-Banking-Software-System (feature-customer-module)
$ git status
On branch feature-customer-module
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
    modified:   Trello/Backlog/Card-10_Customer_View_Account.txt

no changes added to commit (use "git add" and/or "git commit -a")

Asus@Mahirs-Workstation MINGW64 /d/AIUB/AIUB 7th(Spring 25-26) Semester/Software Engineering/Traditional-M
odel-Driven-based-Banking-Software-System (feature-customer-module)
$
```

Figure 32: Customer Module Git Workflow (Part 1)



```
Asus@Mahirs-Workstation MINGW64 /d/AIUB/AIUB 7th(Spring 25-26) Semester/Software Engineering
$ git clone https://github.com/s21gem/Traditional-Model-Driven-based-Banking-Software-System.git
Cloning into 'Traditional-Model-Driven-based-Banking-Software-System'...
remote: Enumerating objects: 30, done.
remote: Counting objects: 100% (30/30), done.
remote: Compressing objects: 100% (29/29), done.
remote: Total 30 (delta 0), reused 30 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (30/30), 7.10 KiB | 201.00 KiB/s, done.

Asus@Mahirs-Workstation MINGW64 /d/AIUB/AIUB 7th(Spring 25-26) Semester/Software Engineering
$ cd Traditional-Model-Driven-based-Banking-Software-System

Asus@Mahirs-Workstation MINGW64 /d/AIUB/AIUB 7th(Spring 25-26) Semester/Software Engineering/Traditional-M
odel-Driven-based-Banking-Software-System (main)
$ git checkout -b feature-customer-module
Switched to a new branch 'feature-customer-module'

Asus@Mahirs-Workstation MINGW64 /d/AIUB/AIUB 7th(Spring 25-26) Semester/Software Engineering/Traditional-M
odel-Driven-based-Banking-Software-System (feature-customer-module)
$ git branch
* feature-customer-module
main

Asus@Mahirs-Workstation MINGW64 /d/AIUB/AIUB 7th(Spring 25-26) Semester/Software Engineering/Traditional-M
odel-Driven-based-Banking-Software-System (feature-customer-module)
$ git status
On branch feature-customer-module
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
    modified:   Trello/Backlog/Card-10_Customer_View_Account.txt

no changes added to commit (use "git add" and/or "git commit -a")
```

Figure 33: Customer Module Git Workflow (Part 2)


```
MINGW64/d/AIUB/AIUB 7th(Spring 25-26) Semester/Software Engineering/Traditional-Model-Driven-based-Bank...
Asus@Mahirs-Workstation MINGW64 /d/AIUB/AIUB 7th(Spring 25-26) Semester/Software Engineering/Traditional-M
odel-Driven-based-Banking-Software-System (feature-customer-module)
$ git add .

Asus@Mahirs-Workstation MINGW64 /d/AIUB/AIUB 7th(Spring 25-26) Semester/Software Engineering/Traditional-M
odel-Driven-based-Banking-Software-System (feature-customer-module)
$ git commit -m "Updated Card-10 acceptance criteria"
[feature-customer-module 8e5fb04] Updated Card-10 acceptance criteria
1 file changed, 2 insertions(+)

Asus@Mahirs-Workstation MINGW64 /d/AIUB/AIUB 7th(Spring 25-26) Semester/Software Engineering/Traditional-M
odel-Driven-based-Banking-Software-System (feature-customer-module)
$ git status
On branch feature-customer-module
nothing to commit, working tree clean

Asus@Mahirs-Workstation MINGW64 /d/AIUB/AIUB 7th(Spring 25-26) Semester/Software Engineering/Traditional-M
odel-Driven-based-Banking-Software-System (feature-customer-module)
$ git push -u origin feature-customer-module
Enumerating objects: 9, done.
Counting objects: 100% (9/9), done.
Delta compression using up to 16 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (5/5), 483 bytes | 483.00 KiB/s, done.
Total 5 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
remote:
remote: Create a pull request for 'feature-customer-module' on GitHub by visiting:
remote:   https://github.com/s21gem/Traditional-Model-Driven-based-Banking-Software-System/pull/new/fea
ture-customer-module
remote:
To https://github.com/s21gem/Traditional-Model-Driven-based-Banking-Software-System.git
 * [new branch]   feature-customer-module -> feature-customer-module
branch 'feature-customer-module' set up to track 'origin/feature-customer-module'.
```

Figure 34: Customer Module Git Workflow (Part 3)

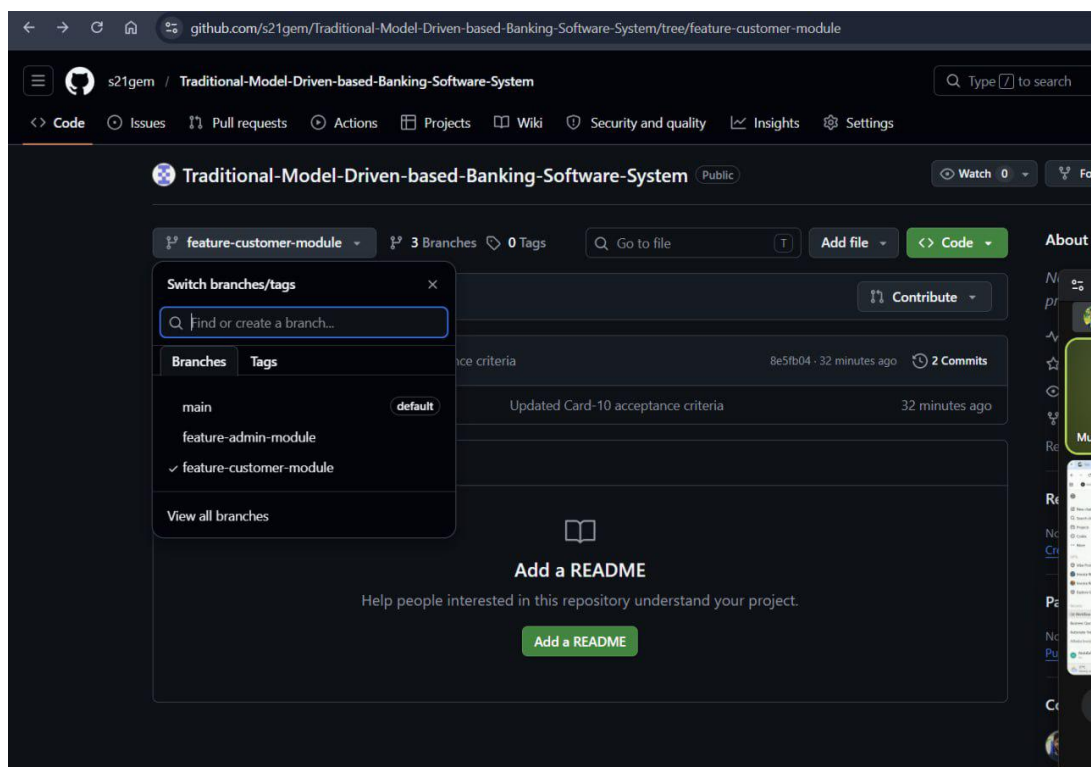


Figure 35: Customer Module Git Workflow (Part 4)

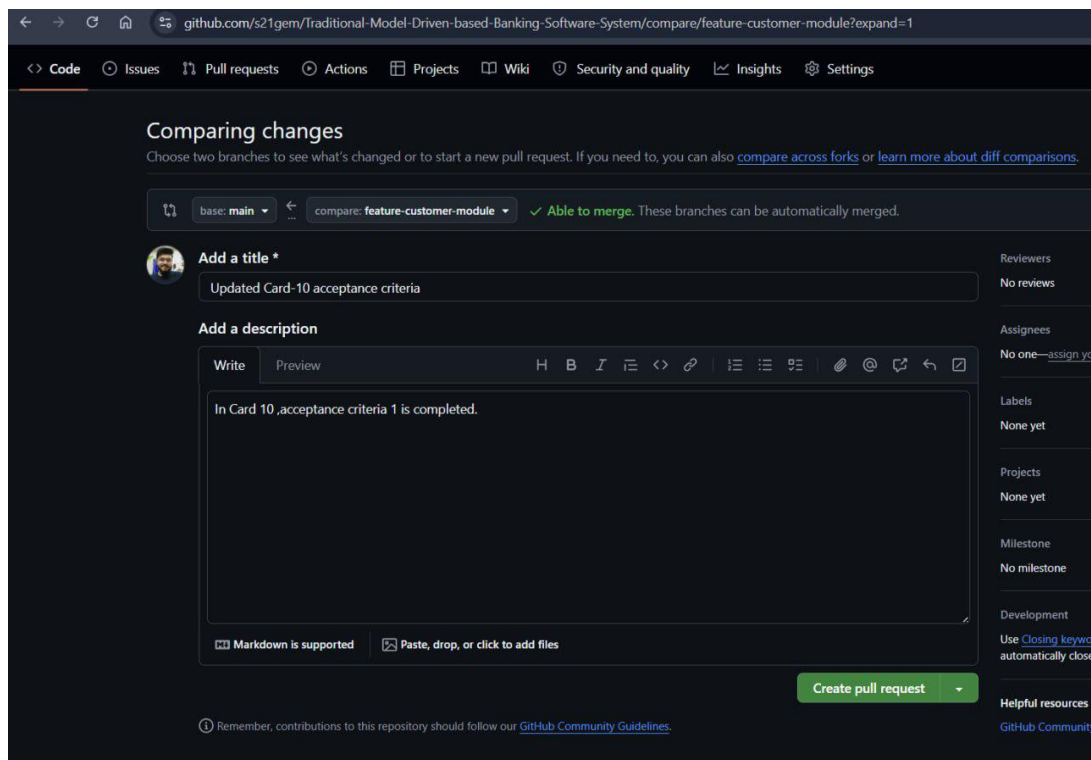


Figure 36: Customer Module Git Workflow (Part 5)

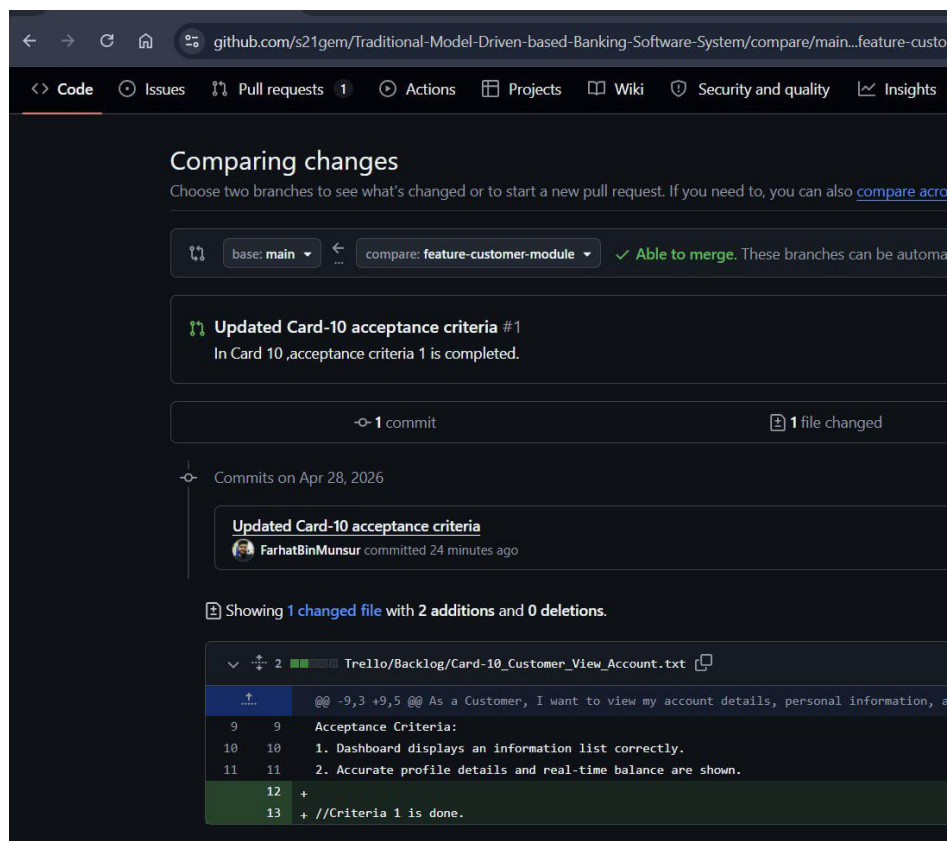


Figure 37: Customer Module Git Workflow (Part 6)

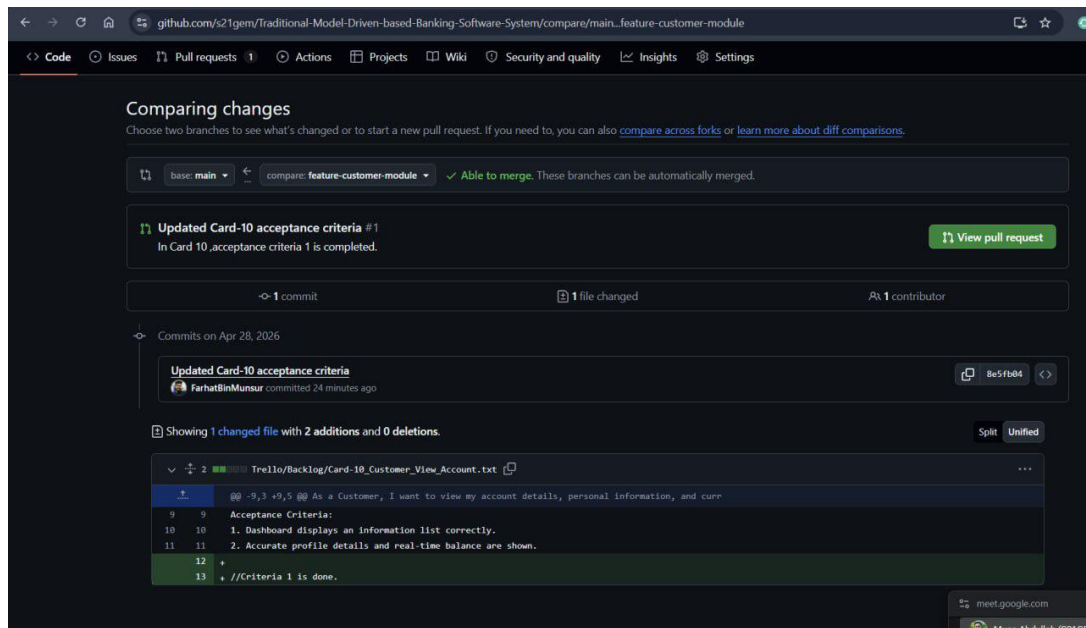


Figure 38: Customer Module Git Workflow (Part 7)

Branch Manager Module Workflow

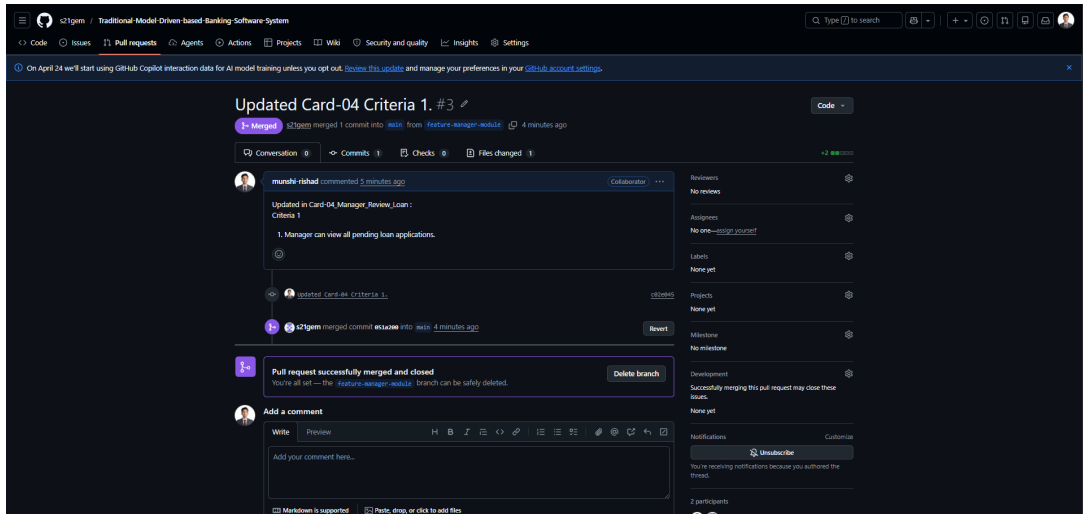


Figure 39: Branch Manager Module Git Workflow (Part 1)

```
MINGW64/c:/Users/Najiat Islam Rishad/Desktop/Traditional-Model-Driven-based-Banking-Software-System
Najiat Islam Rishad@DESKTOP-N9RC7F0 MINGW64 ~/Desktop
$ git clone https://github.com/s2lgem/Traditional-Model-Driven-based-Banking-Software-System.git
Cloning into 'Traditional-Model-Driven-based-Banking-Software-System'...
remote: Enumerating objects: 40, done.
remote: Counting objects: 100% (40/40), done.
remote: Compressing objects: 100% (34/34), done.
remote: Total 40 (delta 3), reused 40 (delta 3), pack-reused 0 (from 0)
Receiving objects: 100% (40/40), 7.94 KiB | 1.58 MiB/s, done.
Resolving deltas: 100% (3/3), done.
Najiat Islam Rishad@DESKTOP-N9RC7F0 MINGW64 ~/Desktop
$ cd Traditional-Model-Driven-based-Banking-Software-System
Najiat Islam Rishad@DESKTOP-N9RC7F0 MINGW64 ~/Desktop/Traditional-Model-Driven-based-Banking-Software-System (main)
$ git checkout -b feature-manager-module
Switched to a new branch 'feature-manager-module'
Najiat Islam Rishad@DESKTOP-N9RC7F0 MINGW64 ~/Desktop/Traditional-Model-Driven-based-Banking-Software-System (feature-manager-module)
$ git branch
* feature-manager-module
  main
Najiat Islam Rishad@DESKTOP-N9RC7F0 MINGW64 ~/Desktop/Traditional-Model-Driven-based-Banking-Software-System (feature-manager-module)
$ git status
On branch feature-manager-module
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   Trello/Backlog/Card-04_Manager_Review_Loan.txt

no changes added to commit (use "git add" and/or "git commit -a")
Najiat Islam Rishad@DESKTOP-N9RC7F0 MINGW64 ~/Desktop/Traditional-Model-Driven-based-Banking-Software-System (feature-manager-module)
$ git add .
Najiat Islam Rishad@DESKTOP-N9RC7F0 MINGW64 ~/Desktop/Traditional-Model-Driven-based-Banking-Software-System (feature-manager-module)
$ git commit -m "Updated Card-04 Criteria 1."
[feature-manager-module c02e045] Updated Card-04 Criteria 1.
1 file changed, 2 insertions(+)
Najiat Islam Rishad@DESKTOP-N9RC7F0 MINGW64 ~/Desktop/Traditional-Model-Driven-based-Banking-Software-System (feature-manager-module)
$ git push -u origin feature-manager-module
remote: Permission to s2lgem/Traditional-Model-Driven-based-Banking-Software-System.git denied to munshi-rishad.
fatal: unable to access 'https://github.com/s2lgem/Traditional-Model-Driven-based-Banking-Software-System.git/': The requested URL returned error: 403
Najiat Islam Rishad@DESKTOP-N9RC7F0 MINGW64 ~/Desktop/Traditional-Model-Driven-based-Banking-Software-System (feature-manager-module)
$ git push -u origin feature-manager-module
Enumerating objects: 9, done.
Counting objects: 100% (9/9), done.
Delta compression using up to 12 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (5/5), 486 bytes | 486.00 KiB/s, done.
Total 5 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
remote: Create a pull request for 'feature-manager-module' on GitHub by visiting:
remote:   https://github.com/s2lgem/Traditional-Model-Driven-based-Banking-Software-System/pull/new/feature-manager-module
To https://github.com/s2lgem/Traditional-Model-Driven-based-Banking-Software-System.git
 * [new branch]      feature-manager-module -> feature-manager-module
branch 'feature-manager-module' set up to track 'origin/feature-manager-module'.
```

Figure 40: Branch Manager Module Git Workflow (Part 2)

General/All User Module Workflow

```
MINGW64/c/Users/alvi/OneDrive/Desktop/Traditional-Model-Driven-based-Banking-Software-System
git config --global user.name "Your Name"

to set your account's default identity.
Omit --global to set the identity only in this repository.

fatal: unable to auto-detect email address (got 'alvif@Alvi_Faris.(none)')

alvif@Alvi_Faris MINGW64 ~/OneDrive/Desktop/Traditional-Model-Driven-based-Banking-Software-System (feature-general-module)
$ AC

alvif@Alvi_Faris MINGW64 ~/OneDrive/Desktop/Traditional-Model-Driven-based-Banking-Software-System (feature-general-module)
$ A[[200~git config --global user.email "23-51449-10student.aiub.edu"
bash: $'\E[200~git': command not found

alvif@Alvi_Faris MINGW64 ~/OneDrive/Desktop/Traditional-Model-Driven-based-Banking-Software-System (feature-general-module)
$ A[[200~$ git config --global user.name "alvifaris-block"
bash: $'\E[200~$': command not found

alvif@Alvi_Faris MINGW64 ~/OneDrive/Desktop/Traditional-Model-Driven-based-Banking-Software-System (feature-general-module)
$ git config --global user.email "23-51449-10student.aiub.edu"

alvif@Alvi_Faris MINGW64 ~/OneDrive/Desktop/Traditional-Model-Driven-based-Banking-Software-System (feature-general-module)
$ git config --global user.name "alvifaris-block"

alvif@Alvi_Faris MINGW64 ~/OneDrive/Desktop/Traditional-Model-Driven-based-Banking-Software-System (feature-general-module)
$ git add .

alvif@Alvi_Faris MINGW64 ~/OneDrive/Desktop/Traditional-Model-Driven-based-Banking-Software-System (feature-general-module)
$ git commit -m "Updated Criteria 1 and 2 in card 18"
[feature-general-module d4d8ec9] Updated Criteria 1 and 2 in card 18
1 file changed, 2 insertions(+).

alvif@Alvi_Faris MINGW64 ~/OneDrive/Desktop/Traditional-Model-Driven-based-Banking-Software-System (feature-general-module)
$ git push -u origin feature-general-module
info: please complete authentication in your browser...
Enumerating objects: 9, done.
Counting objects: 100% (9/9), done.
Delta compression using up to 16 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (5/5), 437 bytes | 437.00 KiB/s, done.
Total 5 (delta 3), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (3/3), completed with 3 local objects.
remote:
remote: Create a pull request for 'feature-general-module' on GitHub by visiting:
remote:   https://github.com/s21gem/Traditional-Model-Driven-based-Banking-Software-System/pull/new/feature-general-module
remote:
to https://github.com/s21gem/Traditional-Model-Driven-based-Banking-Software-System.git
* [new branch] feature-general-module -> feature-general-module
branch 'feature-general-module' set up to track 'origin/feature-general-module'.
```

Figure 41: General/All User Module Git Workflow (Part 1)

```
MINGW64/c/Users/alvi/OneDrive/Desktop/Traditional-Model-Driven-based-Banking-Software-System

alvif@Alvi_Faris MINGW64 ~/OneDrive/Desktop
$ git clone https://github.com/s21gem/Traditional-Model-Driven-based-Banking-Software-System.git
Cloning into 'Traditional-Model-Driven-based-Banking-Software-System'...
remote: Enumerating objects: 46, done.
remote: Counting objects: 100% (46/46), done.
remote: Compressing objects: 100% (37/37), done.
remote: Total 46 (delta 5), reused 45 (delta 5), pack-reused 0 (from 0)
Receiving objects: 100% (46/46), 9.22 KiB | 4.61 MiB/s, done.
Resolving deltas: 100% (5/5), done.

alvif@Alvi_Faris MINGW64 ~/OneDrive/Desktop
$ A[[200~
bash: $'\E[200~': command not found

alvif@Alvi_Faris MINGW64 ~/OneDrive/Desktop
$ cd Traditional-Model-Driven-based-Banking-Software-System

alvif@Alvi_Faris MINGW64 ~/OneDrive/Desktop/Traditional-Model-Driven-based-Banking-Software-System (main)
$ git checkout -b feature-general-module
Switched to a new branch 'feature-general-module'

alvif@Alvi_Faris MINGW64 ~/OneDrive/Desktop/Traditional-Model-Driven-based-Banking-Software-System (feature-general-module)
$ git branch
* feature-general-module
  main

alvif@Alvi_Faris MINGW64 ~/OneDrive/Desktop/Traditional-Model-Driven-based-Banking-Software-System (feature-general-module)
$ git status
On branch feature-general-module
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
    modified:   Trellio/To Do/Card-18-AllUsers_Login_System.txt

no changes added to commit (use "git add" and/or "git commit -a")

alvif@Alvi_Faris MINGW64 ~/OneDrive/Desktop/Traditional-Model-Driven-based-Banking-Software-System (feature-general-module)
$ git add .

alvif@Alvi_Faris MINGW64 ~/OneDrive/Desktop/Traditional-Model-Driven-based-Banking-Software-System (feature-general-module)
$ git commit -m "Updated Criteria 1 and 2 in card 18"
Author identity unknown

*** Please tell me who you are.

Run

git config --global user.email "you@example.com"
git config --global user.name "Your Name"

to set your account's default identity.
Omit --global to set the identity only in this repository.
```

Figure 42: General/All User Module Git Workflow (Part 2)

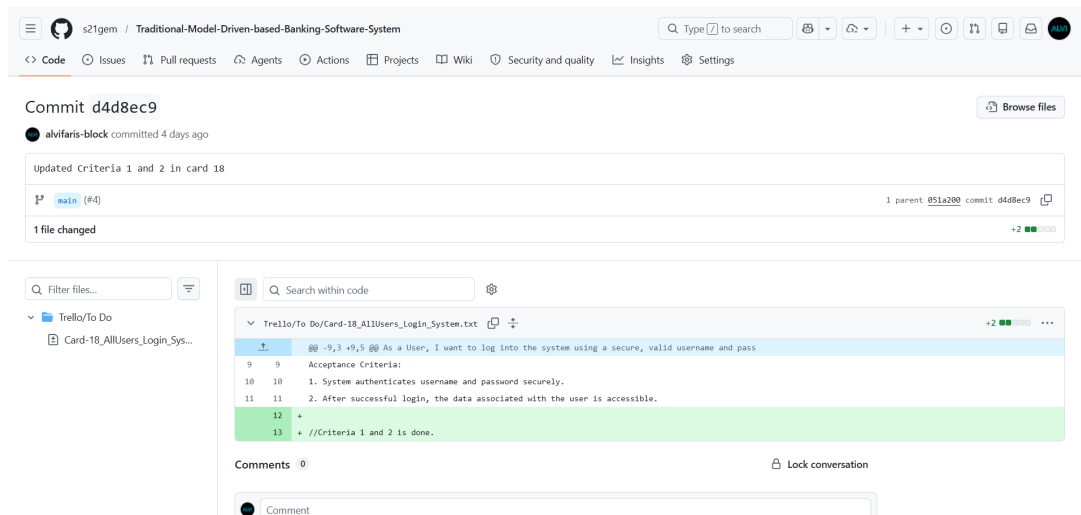


Figure 43: General/All User Module Git Workflow (Part 3)

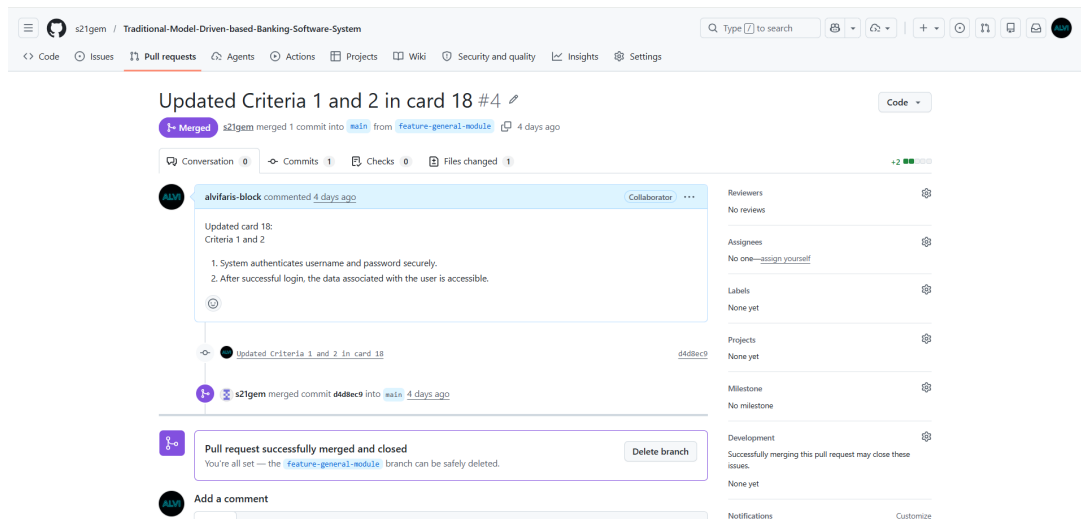


Figure 44: General/All User Module Git Workflow (Part 4)

Final Project Management Board

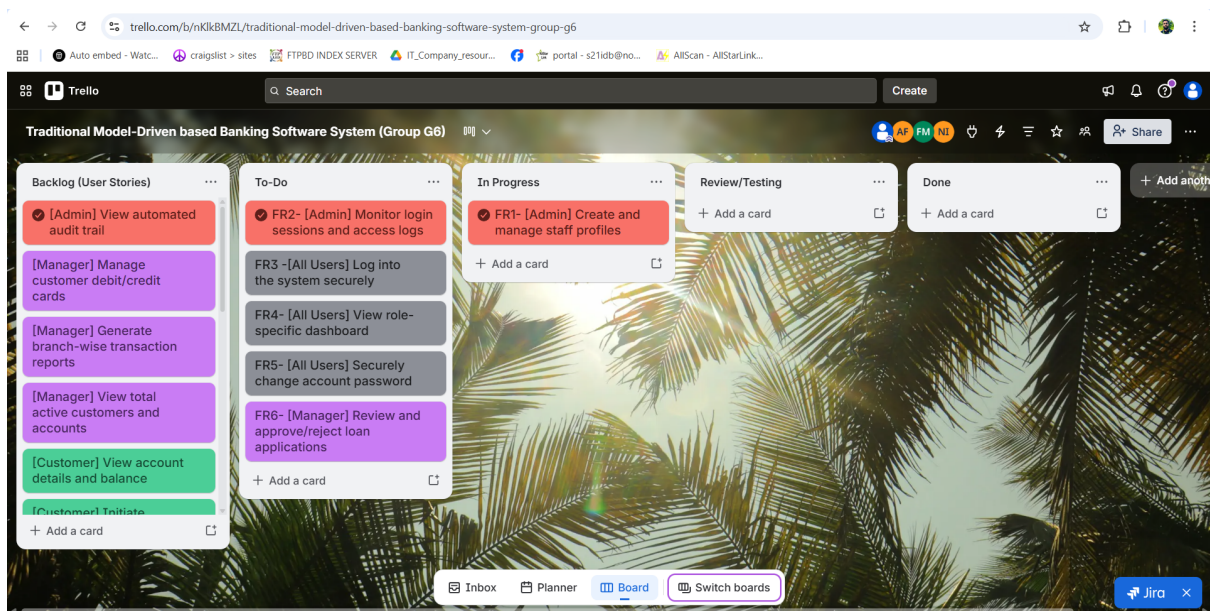


Figure 45: Final Trello Board showing Agile Task Management

6. CONCLUSION

The development of the Traditional Model-Driven based Banking Software System represents a significant step towards modernizing financial operations and enhancing the overall user experience for both bank personnel and customers. By strictly adhering to the V-Model process framework, this project successfully prioritized rigorous verification and validation at every development phase, ensuring a highly secure, scalable, and reliable platform. The comprehensive Software Requirements Specification (SRS) laid a solid foundation, clearly defining the scopes, user stories, and traceability matrices necessary to align the system's functionalities with the precise needs of Administrators, Branch Managers, Tellers, and Customers.

Furthermore, the integration of structured system design using Use Case diagrams provided a clear architectural blueprint, while the Figma wireframes mapped out an intuitive, user-centric interface. The systematic adoption of Git for version control, combined with Trello for Agile task management, facilitated seamless collaboration, transparent tracking, and efficient issue resolution throughout the project lifecycle. In conclusion, this well-documented and methodically executed project not only resolves traditional banking inefficiencies but also establishes a robust, future-proof framework capable of accommodating evolving financial technologies and growing customer demands with unparalleled security and efficiency.